SQL Server AlwaysOn

Gianluca Hotz

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http://www.ugiss.org

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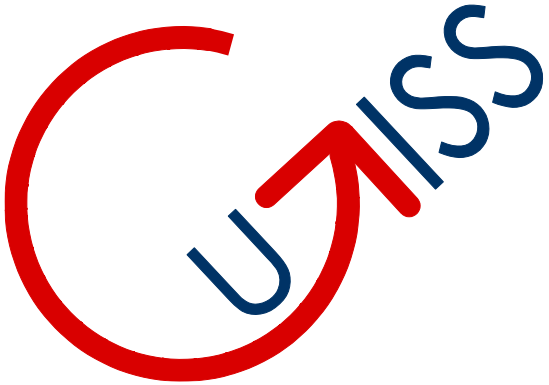


Speaker

▶ Gianluca Hotz

- Consultant and mentor at SolidQ
  - Administration, capacity planning, data modeling, development, optimization
  - Almost 20 years on SQL Server (from 4.21 in 1996)
- Interests
  - Relational model, DBMS architectures, high availability and disaster recovery
- Microsoft MVP SQL Server since 1998
- Founder & vice-president UGISS
  - User Group Italiano SQL Server
  - Official Italian PASS Chapter
- Mail: ghotz@ugiss.org - ghotz@solidq.com

Agenda

- Introduction
- AlwaysOn FCI
- AlwaysOn AG
- AlwaysON Advanced Scenarios

AlwaysOn

▶ Introduction

- New name for HA and DR technologies
  - Failover Cluster Instance (FCI)
  - Availability Groups (AG)
- For the purpose of this presentation
  - High Availability (local HA): in a data center
  - Disaster Recovery: across data centers

AlwaysOn FCI

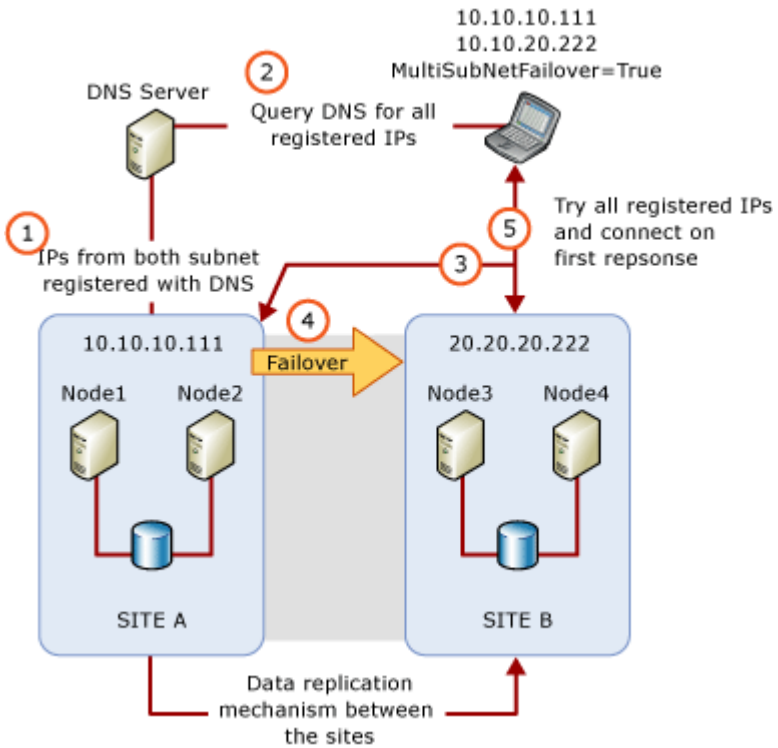
▶ FCI Brief review

- Protects an entire SQL Server instance
- Requires Shared Storage
- Geographic DR with SQL Server < 2012
  - Proprietary solutions for Stretch V-LAN
  - Proprietary solutions for storage replication
- Multi-subnet support >=2012

AlwaysOn FCI

▶ Multi-Subnet Support

- Nodes on different subnets
  - IP Address resources in OR
- Still needs data replication
- Requires Enterprise Edition

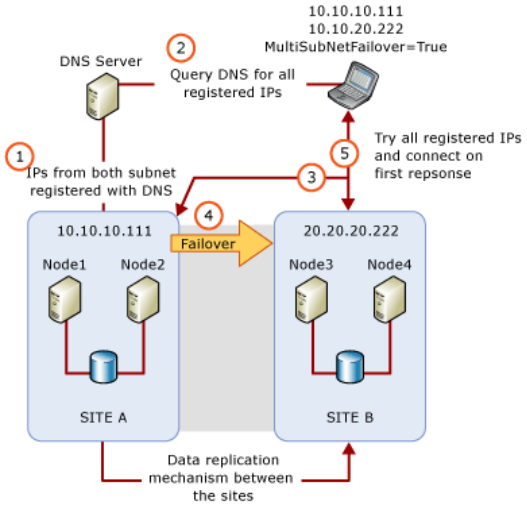


http://msdn.microsoft.com/en-us/library/ff878716.aspx

AlwaysOn FCI

▶ Multi-Subnet Support: how do clients reconnect?

- New clients
  - Try IP addresses in order
  - Avoid DNS update latencies
  - New connection string parameter
    - MultiSubnetFailover=True
    - Try IP addresses in parallel
  - Check driver support!
- Old clients
  - Change timeout value
  - Reduce DNS updates latency (HostRecordTTL)
  - Reduce DNS cache expiration on the client



http://msdn.microsoft.com/en-us/library/ff878716.aspx

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▶ Failover policies in SQL Server <= 2008

- Looks Alive
  - Checks service state (Service Control Mgr.)
  - Every 5 seconds
- IsAlive
  - Checks `SELECT @@SERVERNAME`
  - Every 60 seconds
  - Can pass check even when not responding to other incoming requests
    - e.g. stuck schedulers

AlwaysOn FCI

▶ Flexible failover policies in SQL Server >= 2012

- IsAlive based on system procedure
  - Resource DLL calls **`sp_server_diagnostic`**
    - Over a dedicated connection
  - Diagnostic information from procedure
    - New instance parameter FailureConditionLevel
    - Allows selecting conditions for failover to occur
  - Timeouts while calling diagnostic procedure
    - New Cluster parameter HealthCheckTimeout

AlwaysOn FCI

▶ Failure condition levels

Level	Condition	Description
0	No automatic failover or restart	Never Failover (system maintenance only)
1	Failover or restart on SQL Server down	Service down
2	Failover or restart on SQL Server unresponsive	Server hanged (sp_server_diagnostics not responding within HealthCheckTimeout setting)
3	Failover or restart on critical SQL Server errors	System unhealthy (sp_server_diagnostics “system error”)
4	Failover or restart on moderate SQL Server errors	Resource unhealthy (sp_server_diagnostics “resource error”)
5	Failover or restart on any qualified failure	QP Unhealthy (sp_server_diagnostics “query_processing error”)

AlwaysOn FCI

▶ Flexible failover policies configuration

- Failover Cluster Manager
- PowerShell
  - `Get-ClusterResource $SqlName`
    - `| Set-ClusterParameter HealthCheckTimeout 60000`
    - `| Set-ClusterParameter FailureConditionLevel 3`
- T-SQL
  - `ALTER SERVER CONFIGURATION SET FAILOVER CLUSTER PROPERTYHEALTHCHECKTIMEOUT = 60000;`
  - `ALTER SERVER CONFIGURATION SET FAILOVER CLUSTER PROPERTY FAILURECONDITIONLEVEL = 3;`

AlwaysOn FCI

▶ Reducing planned downtime with SQL Server >= 2012

- Windows Core Support
  - Less patches to be installed (circa 50-60%)
- Support for rolling upgrading and patching
- Better control on failover time
  - Indirect Checkpoints
    - `ALTER DATABASE …`
      `SET TARGET_RECOVERY_TIME = …`

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▶ New Storage options with SQL Server >= 2012

- Tempdb on local storage
  - Can leverage fast SSD cards
- SMB shares supported
  - Best on
    - Windows Server 2008 R2 (SMB 2.1 + Hotfix)
    - Windows Server 2012
    - Azure
- Asymmetric Disks

AlwaysOn FCI

▶ Failover Cluster Instances in SQL Server >= 2014

- Support for Windows Server 2012 CSV
- Support for FCI in Sysprep
- Same support as for Availability Groups in
  - `sys.dm_hadr_cluster`
  - `sys.dm_hadr_cluster_members`
  - `sys.dm_hadr_cluster_networks`
- New DMV
  - `sys.dm_io_cluster_valid_path_names`

AlwaysOn AG

▶ Data replicas review

- Log Shipping
- Replication
- Database Mirroring
- Availability Groups
  - >= SQL Server 2012

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▶ Key evaluation factors for data replication

- Granularity
  - Instance, database, table, subset of table…
- Reliability
  - Number of replicas, synchronous…
- Performance
  - Asynchronous, off-load activities
- Manageability
  - Setup, management, number of moving parts…
- Failover transparency
  - Manual, automatic, coordinated between parts…

AlwaysOn AG

▶ Log Shipping

- Advantages
  - Entire database replicas
  - Multiple replicas per database
  - Simple and robust
- Disadvantages
  - Only asynchronous replicas
  - Readable replicas with trade-offs
    - Updated to last log restore
    - Need to disconnect clients before restoring

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▶ Replication

- Advantages
  - More granular (depends on scenario)
  - Multiple replicas
  - Readable & updateable replicas
- Disadvantages
  - More granular (depends on scenario)
  - Only asynchronous replicas
  - Typically higher latency
  - Complexity and maintenance overhead

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▶ Database Mirroring

- Advantages
  - Entire database replicas
  - Synchronous and asynchronous (EE) replicas
  - Simple and robust
- Disadvantages
  - Only one replica per database
  - Readable replicas need Database Snapshot
    - Enterprise Edition
    - Updated to last snapshot creation time
    - Need to disconnect or handle multiple snapshot names

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▶ Common Problems

- Manual Failover*
- No coordination to failover multiple DBs
- No backups on replicas*
- Specific limitations
  - E.g. FILESTREAM, cross-database consistency of transactions, only one replica

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▶ AG as evolution of Database Mirroring

- Coordinated failover of multiple DBs
- Multiple replicas (up to 4 in 2012 and 8 in 2014)
- Synchronous/asynchronous replicas
- Up-to-date readable replicas
- Ability to off-load maintenance on replicas e.g. backups

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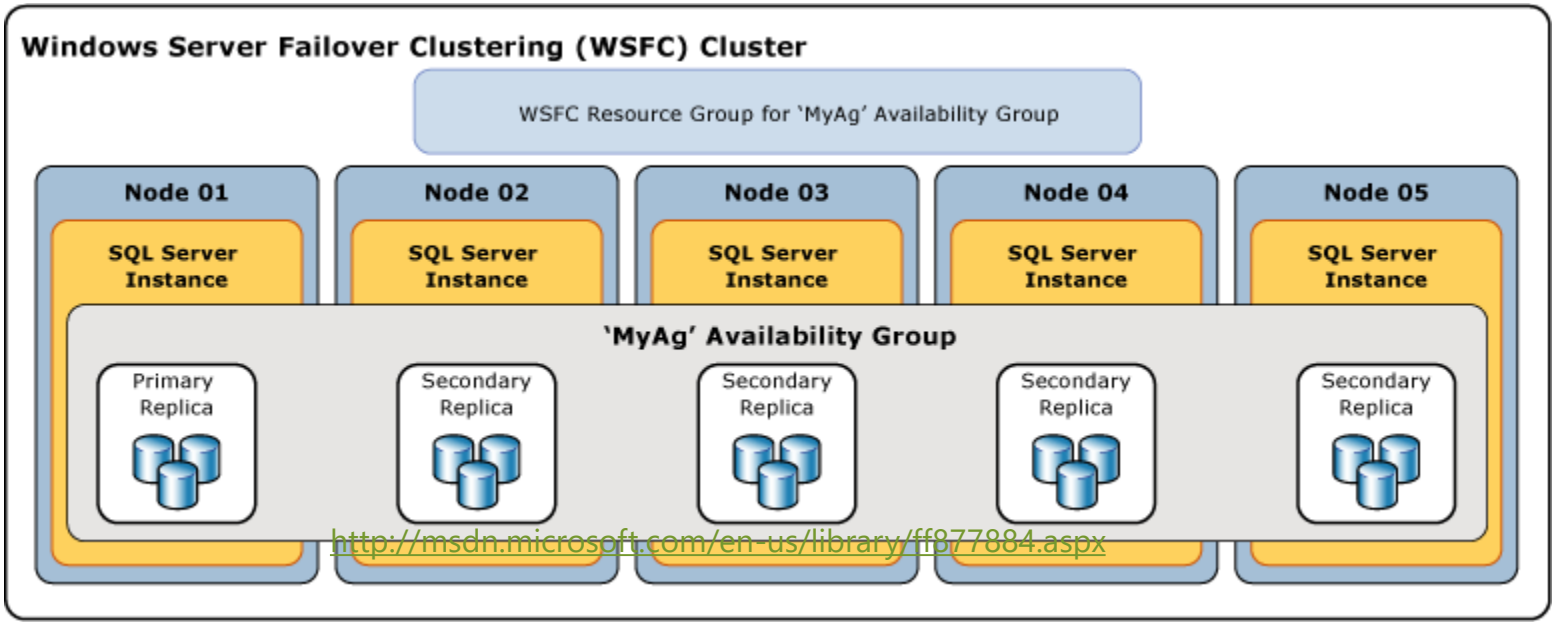
▶ AG Overview



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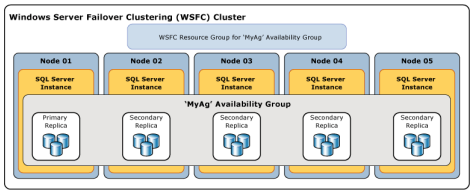
▶ Windows Server Failover Clustering (WSFC)



- Used by Availability Groups for
  - Primary health detection
  - Inter-node health detection
  - Failover coordination
  - Distributed data store for settings and state
  - Distributed change notifications between nodes

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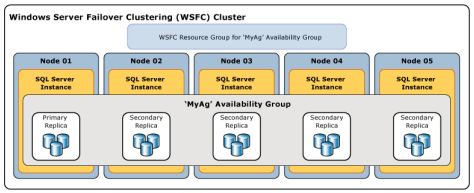
▶ Groups and databases



- Availability Group
  - Set of availability databases failover over together
- Availability Database
  - Database belonging to an availability group
- Primary Database
  - Read/write copy of an availability database
- Secondary Database
  - Read only copy of an availability database

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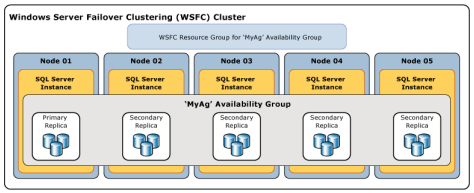
▶ Replicas



- Availability Replica
  - Instance of availability group hosted on a specific SQL Server instance
- Primary Replica
  - Makes primary databases available in read/write
  - Send log records to synchronize secondary replicas
- Secondary Replica
  - Maintains read/only copies of availability DBs

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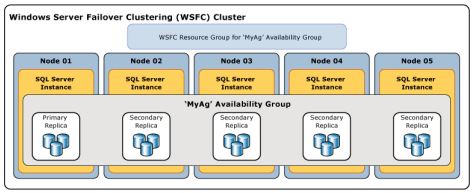
▶ Connectivity



- Availability Group Listener
  - Virtual server name to access replicas
  - Managed by WSFC
- Mirroring Endpoint
  - Used to send log records
  - Synchronization mechanism similar to Database Mirroring

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▶ Availability Group Listener

- Defined by
  - Virtual Network Name (VNN)
  - TCP Port
  - One or more IP addresses (static or DHCP)
- Client connection
  - Application Intent allows automatic routing
  - Support for Mirroring connection string
  - SPN registration automatic for Kerberos auth.

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▶ Synchronization and Failover

- Availability Mode
  - Synchronous
  - Asynchronous
- Failover mode
  - Automatic
  - Manual
  - Forced

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▶ Conditions for Automatic Failover

- Primary and Secondary need
  - Synchronous Availability Mode configuration
  - Automatic Failover configuration
  - To be synchronized
- WFSC needs to reach a quorum
- Failover policies met
  - Leverages FCI mechanism

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▶ Flexible failover policies for Availability Groups

- Leverages FCI mechanism
  - `sp_server_diagnostic`
- Configuration in T-SQL
  - `ALTER AVAILABILITY GROUP …`
    `SET (HEALTH_CHECK_TIMEOUT = 30000);`
  - `ALTER AVAILABILITY GROUP …`
    `SET (FAILURE_CONDITION_LEVEL = 3);`

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▶ Failure condition levels for Availability Groups

- On server down
- On server unresponsive
- On critical server error
- On moderate server error
- On any qualified failure conditions

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▶ Replicas and backups

- Backup support on secondary replicas
  - Full backups with COPY_ONLY
  - Differential not supported!
  - Log backups
    - Form a single log-chain across all replicas
- Automated Backup Preference & Priority
  - No immediate effect
  - Backup scripts need to query UDF
    - `sys.fn_hadr_backup_is_preferred_replica('dbname')`

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▶ Read-Only Access

- Replicas roles
  - Primary: READ_WRITE | ALL
  - Secondary: NO | READ_ONLY | ALL
- New connection property
  - ApplicationIntent
  - Name slightly different across client APIs
- Read-Only Routing

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▶ Topology example 1

Direct attached storage local, regional and geo secondaries

Synchronous data movement

Asynchronous data movement

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▶ Read-Only Access Capacity Planning

- Synchronization latency of replica can raise because of the read I/O activity
- Impact on tempdb
  - Temporary statistics
    - Automatically created
    - Gone after failover
  - Read Committed Snapshot Isolation
    - 14 bytes added to rows at primary (on updates)

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▶ Problem with replicas at database level

- Need Manual synchronization for objects outside database scope
  - New logins/users
  - Jobs
- Solution for logins/users
  - Contained databases
  - Problem with ApplicationIntent=ReadOnly

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▶ Contained Database

- Allows databases isolation
  - Remove dependecies from instance objects
  - It's a databases level property
  - Only partial support in SQL Server 2012
    - DMV/Extended Event to find non contained entities
- Main usage: database level authentication
  - Login not needed, user is used instead
  - Needs to be permitted at instance level

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▶ Interoperability with Analysis Services

- No support for native storage
  - Common pattern: NLB and custom replica
  - Official architectural guide from 2008 http://download.microsoft.com/download/A/5/7/A575AD7C-4172-42D0-8D58-0698D6802F81/SSASReadOnlyDBs.docx
- In general, data sources need to point to listener
- Workload off-loading needs to specify ApplicationIntent

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▶ Interoperability with Reporting Services

- Support for database used by reports
  - Data sources point to listener
  - Data sources may specify ApplicationIntent
- Limited support for RS databases
  - No automatic fail-over
  - Common pattern:
    scale-out deployment with FCI and AG

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▶ Interoperability with SSIS Catalog

- No support out-of-the-box
- Needs custom failover procedure
  - http://blogs.msdn.com/b/mattm/archive/2012/09/19/ssis-with-alwayson.aspx

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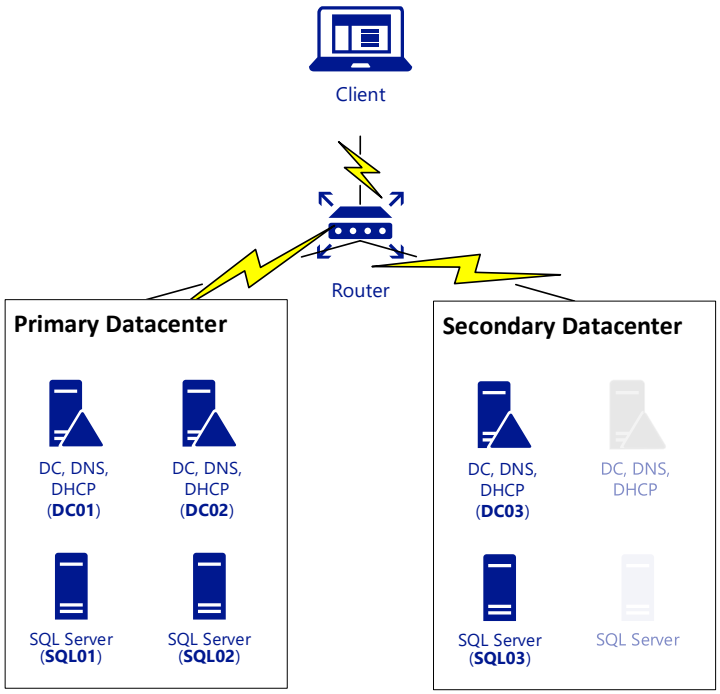
▶ Demo

- Primary Datacenter
  - Primary replica
  - Secondary replica for HA and reporting
- Secondary Datacenter
  - Secondary replica for DR



Step-by-step AG

- Cluster configuration before starting: network primary datacenter

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
- SQLCLUSTER.hadr.local
  - Roles
  - Nodes
  - Storage
    - Disks
    - Pools
  - Networks
  - Cluster Events

Networks (2)

Search | Queries

Name	Status	Cluster Use	Information
Primary Datacenter Network	Up	Cluster and Client	
Secondary Datacenter Network	Up	Cluster and Client	

Primary Datacenter Network

Subnets: 10.0.0.0/24

Summary | Network Connections

Actions

Networks
- Live Migration Settings...
- View
- Refresh
- Help

Primary Datacenter Network
- Information Details...
- Show Critical Events
- Properties
- Help

Networks: Primary Datacenter Network

ITA 2:18 PM

Step-by-step AG

- Cluster configuration before starting: network secondary datacenter

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
- SQLCLUSTER.hadr.local
  - Roles
  - Nodes
  - Storage
    - Disks
    - Pools
  - Networks
  - Cluster Events

Networks (2)

Search | Queries

Name	Status	Cluster Use	Information
Primary Datacenter Network	Up	Cluster and Client	
Secondary Datacenter Network	Up	Cluster and Client	

Secondary Datacenter Network

Subnets: 10.0.1.0/24

Summary | Network Connections

Actions

Networks
- Live Migration Settings...
- View
- Refresh
- Help

Secondary Datacenter Network
- Information Details...
- Show Critical Events
- Properties
- Help

ITA 2:21 PM

Step-by-step AG

- Cluster configuration before starting: summary

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
- SQLCLUSTER.hadr.local
  - Roles
  - Nodes
  - Storage
    - Disks
    - Pools
  - Networks
  - Cluster Events

Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUSTER

SQLCLUSTER has 0 clustered roles and 3 nodes.

Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWitness)
Networks: Primary Datacenter Network, Secondary Datacenter Network
Subnets: 2 IPv4 and 0 IPv6

Configure

Navigate

Cluster Core Resources

Name	Status	Information
Server Name		
Name: SQLCLUSTER	Online	
IP Address: 10.0.0.5	Online	
IP Address: 10.0.1.5	Offline	
File Share Witness		
File Share Witness (\\RRAS\SQLClusterWitness)	Online	

Actions

SQLCLUSTER.hadr.local
- Configure Role...
- Validate Cluster...
- View Validation Report
- Add Node...
- Close Connection
- Reset Recent Events
- More Actions
- View
- Refresh
- Properties
- Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: network name general

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
Roles
Nodes
Storage
Disks
Pools
Networks
Cluster Events

Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUSTER
SQLCLUSTER has 0 clustered roles and
Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWit

Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

Name: SQLCLUSTER Properties

General | Dependencies | Policies | Advanced Policies

Name: Cluster Name
Type: Network Name
Status: Online

Publish PTR records

DNS Name:
SQLCLUSTER

Full Name: SQLCLUSTER.hadr.local

IP Addresses:

Network	IP Address
10.0.0.0/24	10.0.0.5
10.0.1.0/24	10.0.1.5

Add Edit Remove

NetBIOS Status: OK
DNS Status: OK
Kerberos Status: OK

OK Cancel Apply

Actions
SQLCLUSTER.hadr.local
Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
Refresh
Properties
Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: network name dependencies

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
Roles
Nodes
Storage
Disks
Pools
Networks
Cluster Events

Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUSTER
SQLCLUSTER has 0 clustered roles and
Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWit

Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

Name: SQLCLUSTER Properties

General | Dependencies | Policies | Advanced Policies

Specify the resources that must be brought online before this resource can be brought online:

AND/OR	Resource
	Cluster IP Address
OR	Cluster IP Address 10.0.1.5
*	Click here to add a dependency

Insert Delete

(Cluster IP Address OR Cluster IP Address 10.0.1.5)

OK Cancel Apply

Actions
SQLCLUSTER.hadr.local
Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
Refresh
Properties
Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: network advanced policies

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
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Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUSTER
SQLCLUSTER has 0 clustered roles and
Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWit

Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

Name: SQLCLUSTER Properties

General | Dependencies | Policies | Advanced Policies

Clear the check box if you do not want a node to host this resource or this clustered instance.

Possible Owners:

- [x] SQL01
- [x] SQL02
- [x] SQL03

Basic resource health check interval

- (•) Use standard time period for the resource type
- ( ) Use this time period (mm:ss): 00:05

Thorough resource health check interval

- (•) Use standard time period for the resource type
- ( ) Use this time period (mm:ss): 01:00

[ ] Run this resource in a separate Resource Monitor

Choose this option if the associated resource type DLL needs to be debugged or is likely to conflict with other resource type DLLs.

OK | Cancel | Apply

Actions
SQLCLUSTER.hadr.local
Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
Refresh
Properties
Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: primary IP address general

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
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Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUST
SQLCLUSTER has 0 clustered roles and
Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWit

Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

IP Address: 10.0.0.5 Properties

General | Dependencies | Policies | Advanced Policies

Name: Cluster IP Address
Type: IP Address
Status: Online

Network: 10.0.0.0/24
Subnet mask: 255.255.255.0

IP Address
DHCP Enabled
Address: 0.0.0.0
Lease Obtained: <not configured>
Lease Expires: <not configured>
Static IP Address
Address: 10 . 0 . 0 . 5

Enable NetBIOS for this address

OK Cancel Apply

Actions
SQLCLUSTER.hadr.local
Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
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Properties
Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: primary IP address advanced policies

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
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Cluster SQLCLUSTER.hadr.local

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Recent Cluster Events: None in the last 2 hours
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Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

IP Address: 10.0.0.5 Properties

General | Dependencies | Policies | Advanced Policies

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- [x] SQL02
- [ ] SQL03

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- ( ) Use this time period (mm:ss): 00:05

Thorough resource health check interval

- (•) Use standard time period for the resource type
- ( ) Use this time period (mm:ss): 01:00

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OK | Cancel | Apply

Actions
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Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
Refresh
Properties
Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: secondary IP address general

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
Roles
Nodes
Storage
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Cluster Events

Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUST
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Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWit

Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

IP Address: 10.0.1.5 Properties

General | Dependencies | Policies | Advanced Policies

Name: Cluster IP Address 10.0.1.5
Type: IP Address
Status: Offline

Network: 10.0.1.0/24
Subnet mask: 255.255.255.0

IP Address
DHCP Enabled
Address: 0.0.0.0
Lease Obtained: <not configured>
Lease Expires: <not configured>
Static IP Address
Address: 10 . 0 . 1 . 5

Enable NetBIOS for this address

OK Cancel Apply

Actions
SQLCLUSTER.hadr.local
Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
Refresh
Properties
Help

ITA 1:44 PM

Step-by-step AG

- Cluster configuration before starting: secondary IP address general

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
SQLCLUSTER.hadr.local
Roles
Nodes
Storage
Disks
Pools
Networks
Cluster Events

Cluster SQLCLUSTER.hadr.local

Summary of Cluster SQLCLUST
SQLCLUSTER has 0 clustered roles and
Name: SQLCLUSTER.hadr.local
Current Host Server: SQL01
Recent Cluster Events: None in the last 2 hours
Witness: File Share Witness (\\RRAS\SQLClusterWit

Configure

Navigate

Cluster Core Resources

Name
Server Name
Name: SQLCLUSTER
IP Address: 10.0.0.5
IP Address: 10.0.1.5
File Share Witness
File Share Witness (\\RRAS\SQLClusterWit

IP Address: 10.0.1.5 Properties

General | Dependencies | Policies | Advanced Policies

Clear the check box if you do not want a node to host this resource or this clustered instance.

Possible Owners:

- [ ] SQL01
- [ ] SQL02
- [x] SQL03

Basic resource health check interval
- (•) Use standard time period for the resource type
- ( ) Use this time period (mm:ss): 00:05

Thorough resource health check interval
- (•) Use standard time period for the resource type
- ( ) Use this time period (mm:ss): 01:00

[ ] Run this resource in a separate Resource Monitor
Choose this option if the associated resource type DLL needs to be debugged or is likely to conflict with other resource type DLLs.

OK | Cancel | Apply

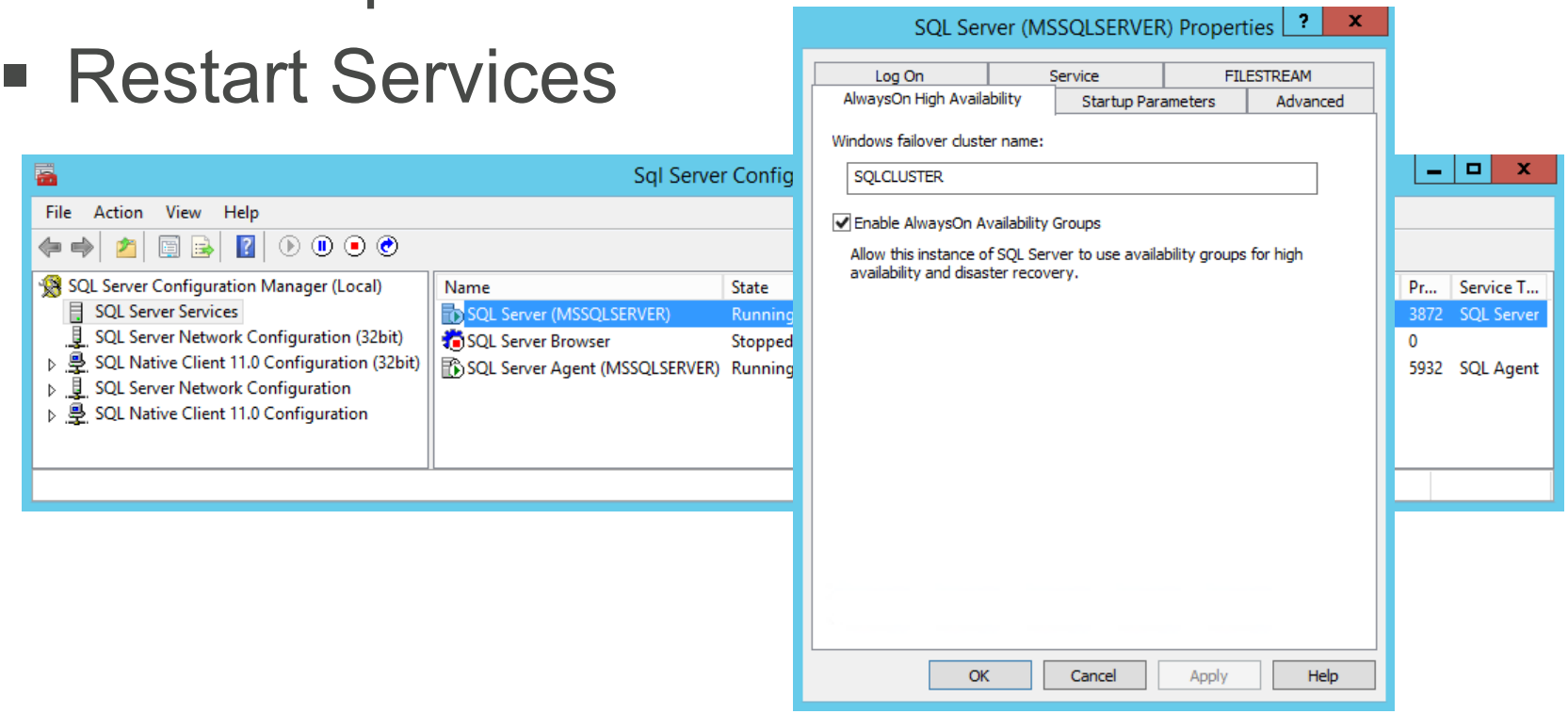
Actions
SQLCLUSTER.hadr.local
Configure Role...
Validate Cluster...
View Validation Report
Add Node...
Close Connection
Reset Recent Events
More Actions
View
Refresh
Properties
Help

ITA 1:44 PM

Step-by-step AG

▶ Enable AlwaysOn

- Enable option
- Restart Services



Step-by-step AG

▶ Enable AlwaysOn with PowerShell

- Use -Force to restart service
- Still need to restart SQL Agent service

Administrator: Windows PowerShell ISE

File Edit View Tools Debug Add-ons Help

ag.01.config.06.set-ag-option-all-nodes.ps1

```
1   Import-Module "sqlps" -DisableNameChecking;
2
3   "SQL01", "SQL02", "SQL03" | % {
4       Write-Output "Enabling AlwaysOn AG on instance $($_)";
5       Enable-sqlAlwaysOn -ServerInstance $_ -Force;
6
7       Write-Output "Starting SQL Server Agent on server $($_)";
8       if ((Get-Service -ComputerName $_ -Name "SQLSERVERAGENT").Status -ne "Running")
9           { (Get-Service -ComputerName $_ -Name "SQLSERVERAGENT").Start(); }
10      while ((Get-Service -ComputerName $_ -Name "SQLSERVERAGENT").Status -ne "Running")
11          { sleep -Seconds 1; }
12
13  }
```

Step-by-step AG

- Create sample databases

ag.02.setup.01.create-databases.sql - SQL01.master (ghotz (58)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQL01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

Object Explorer | Registered Servers

ag.02.setup.01.crea....master (ghotz (58)) | ag.02.setup.02.back....master (ghotz (57))

```sql
IF DB_ID('AGDB01') IS NOT NULL
    DROP DATABASE AGDB01;
GO
CREATE DATABASE AGDB01;
GO
IF DB_ID('AGDB02') IS NOT NULL
    DROP DATABASE AGDB02;
GO
CREATE DATABASE AGDB02;
GO

CREATE TABLE AGDB01.dbo.T1 (F1 int not null PRIMARY KEY);
GO
INSERT  AGDB01.dbo.T1
VALUES  (1), (2), (3), (4), (5);
GO
CREATE TABLE AGDB02.dbo.T2 (F1 int not null PRIMARY KEY);
GO
INSERT  AGDB02.dbo.T2
VALUES  (1), (2), (3), (4), (5);
GO
```

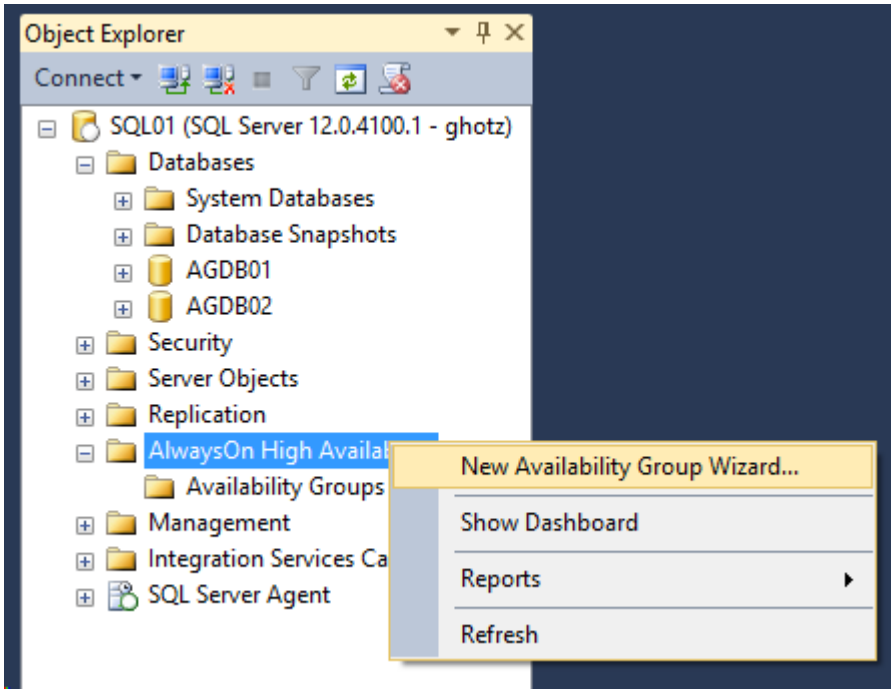
100 %

Query executed successfully. | SQL01 (12.0 SP1) | ghotz (58) | master | 00:00:00 | 0 rows

Ready | Ln 16 | Col 3 | Ch 3 | INS

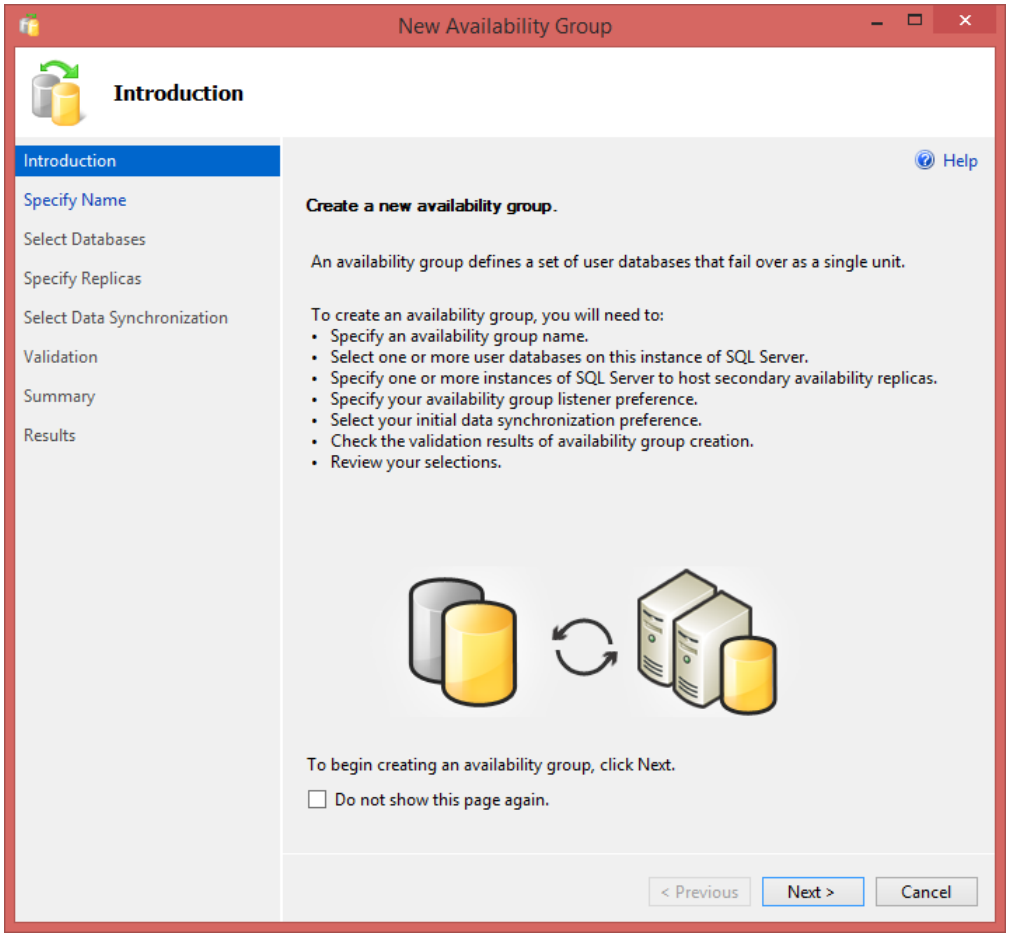
Step-by-step AG

▶ Select Availability Groups Wizard



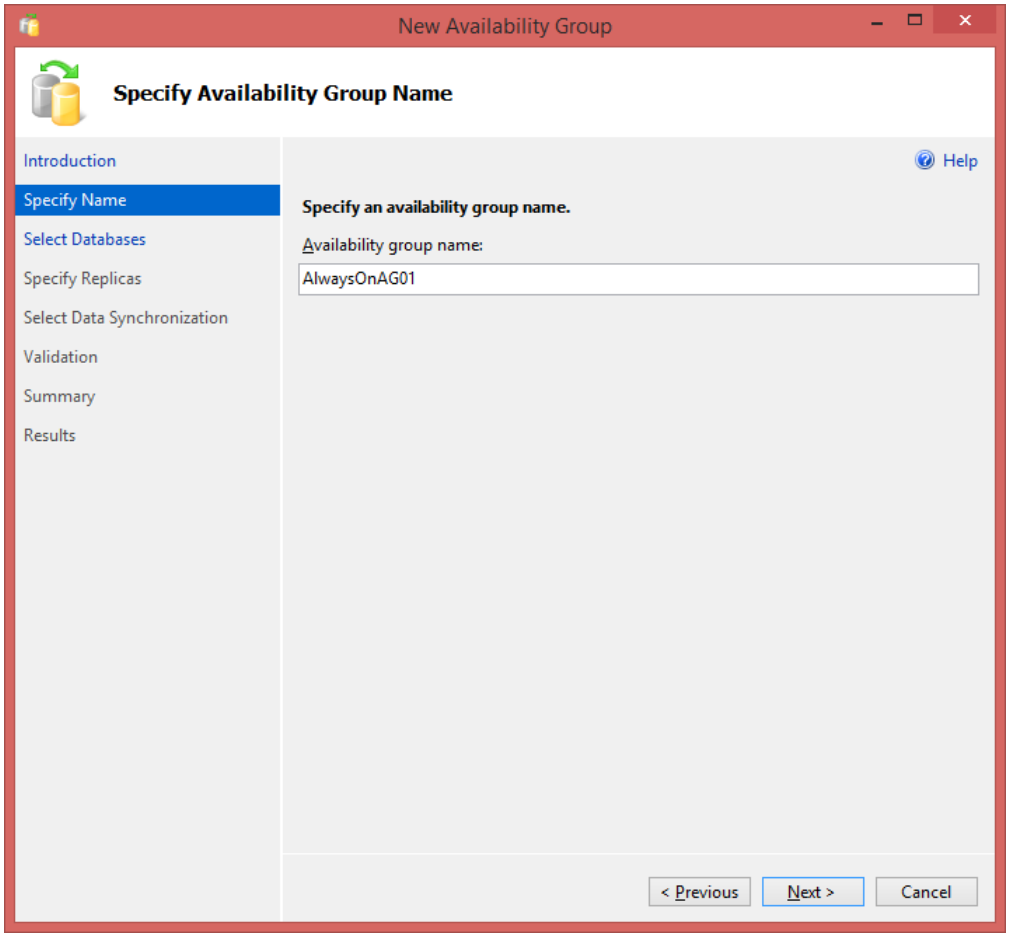
Step-by-step AG

▶ AG Wizard: Start



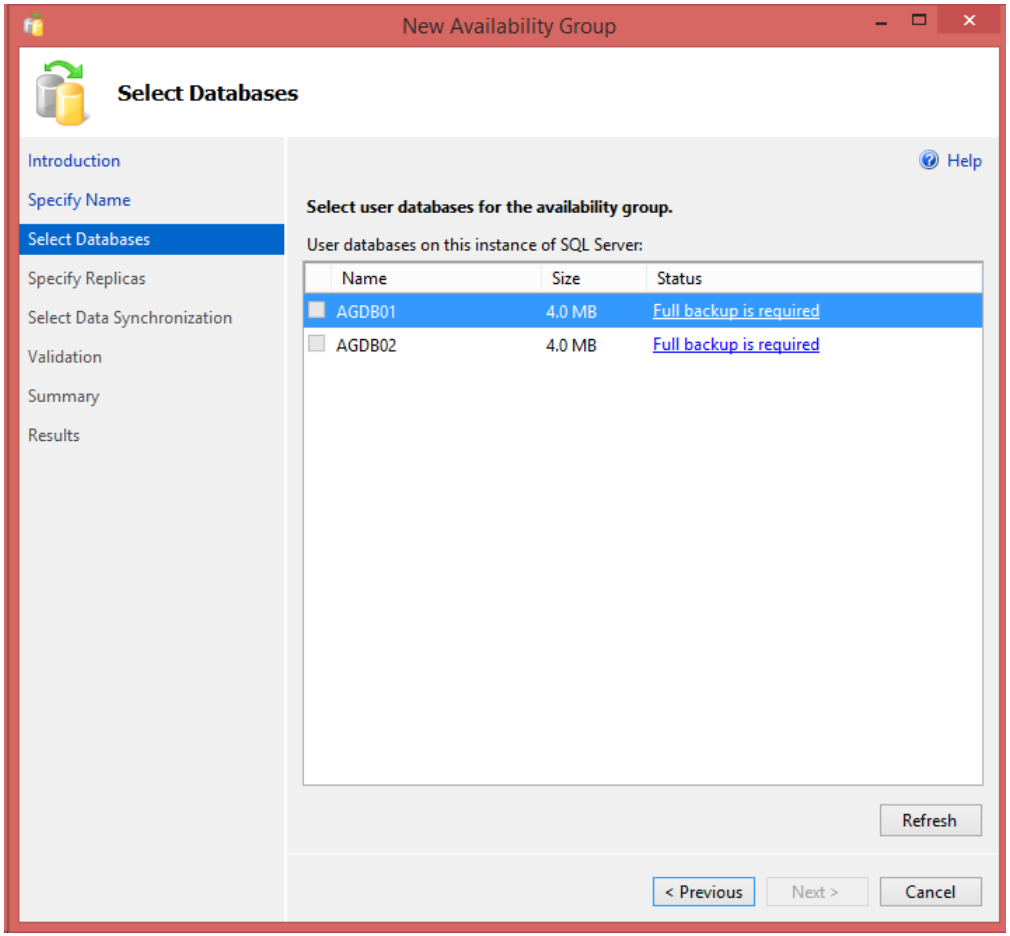
Step-by-step AG

▶ AG Wizard: Assign Name to Group



Step-by-step AG

- AG Wizard: full backup required to choose a database



Step-by-step AG

- AG Wizard: execute full backup

ag.02.setup.02.backup-databases.sql - SQL01.master (ghotz (58)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQL01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

ag.02.setup.02.back....master (ghotz (58))

```
BACKUP DATABASE AGDB01 TO DISK = '\\RRAS\SQLBackups\AGDB01.bak' WITH INIT;
BACKUP DATABASE AGDB02 TO DISK = '\\RRAS\SQLBackups\AGDB02.bak' WITH INIT;
GO
```

100 %

Messages

```
Processed 296 pages for database 'AGDB01', file 'AGDB01' on file 1.
Processed 7 pages for database 'AGDB01', file 'AGDB01_log' on file 1.
BACKUP DATABASE successfully processed 303 pages in 0.094 seconds (25.141 MB/sec).
Processed 296 pages for database 'AGDB02', file 'AGDB02' on file 1.
Processed 7 pages for database 'AGDB02', file 'AGDB02_log' on file 1.
BACKUP DATABASE successfully processed 303 pages in 0.056 seconds (42.201 MB/sec).
```

100 %

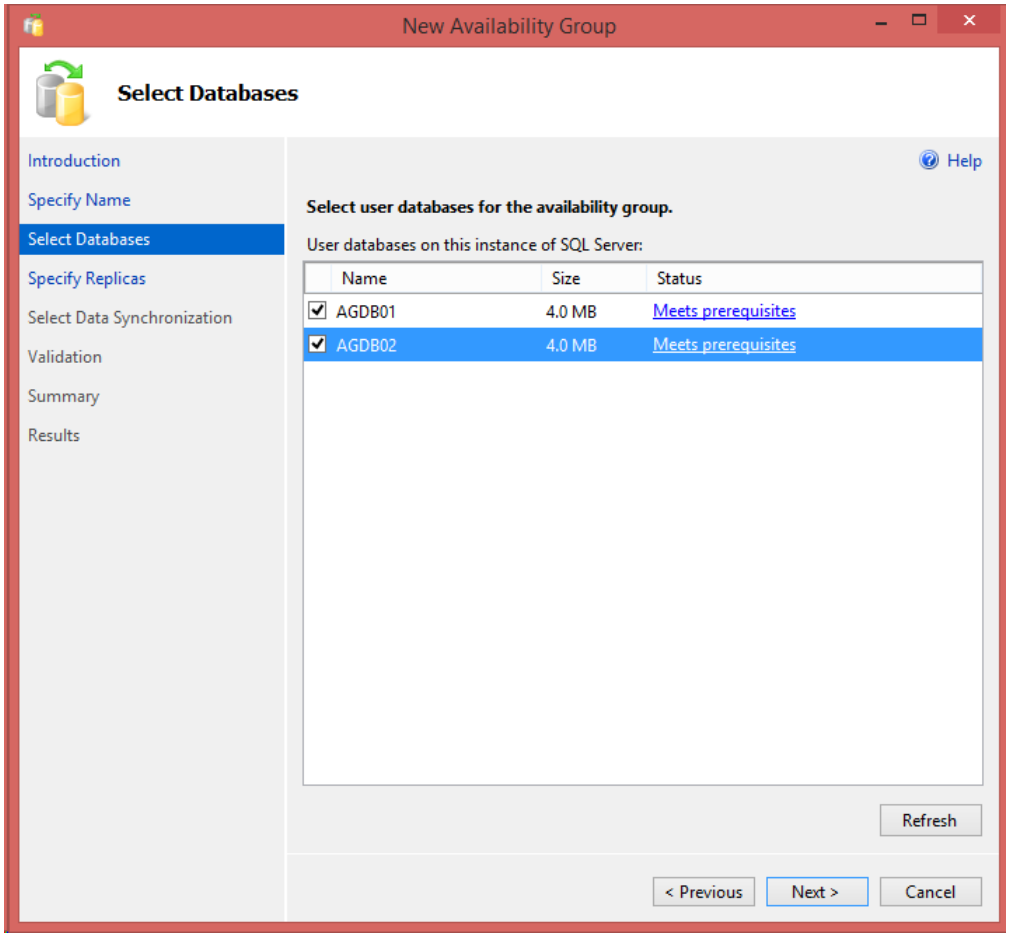
Query executed successfully. | SQL01 (12.0 SP1) | ghotz (58) | master | 00:00:00 | 0 rows

Object Explorer | Registered Servers

Ready | Ln 4 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ AG Wizard: Select databases



Step-by-step AG

▶ AG Wizard: configure Replicas

New Availability Group

Specify Replicas

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Availability Replicas:

Server Instance	Initial Role	Automatic Failover (Up to 2)	Synchronous Commit (Up to 3)	Readable Secondary
SQL01	Primary	☑	☑	No
SQL02	Secondary	☑	☑	No
SQL03	Secondary	☐	☐	No

Add Replica... | Add Azure Replica... | Remove Replica

Summary for the replica hosted by SQL02

Replica mode: Synchronous commit with automatic failover
This replica will use synchronous-commit availability mode and support both automatic failover and manual failover.

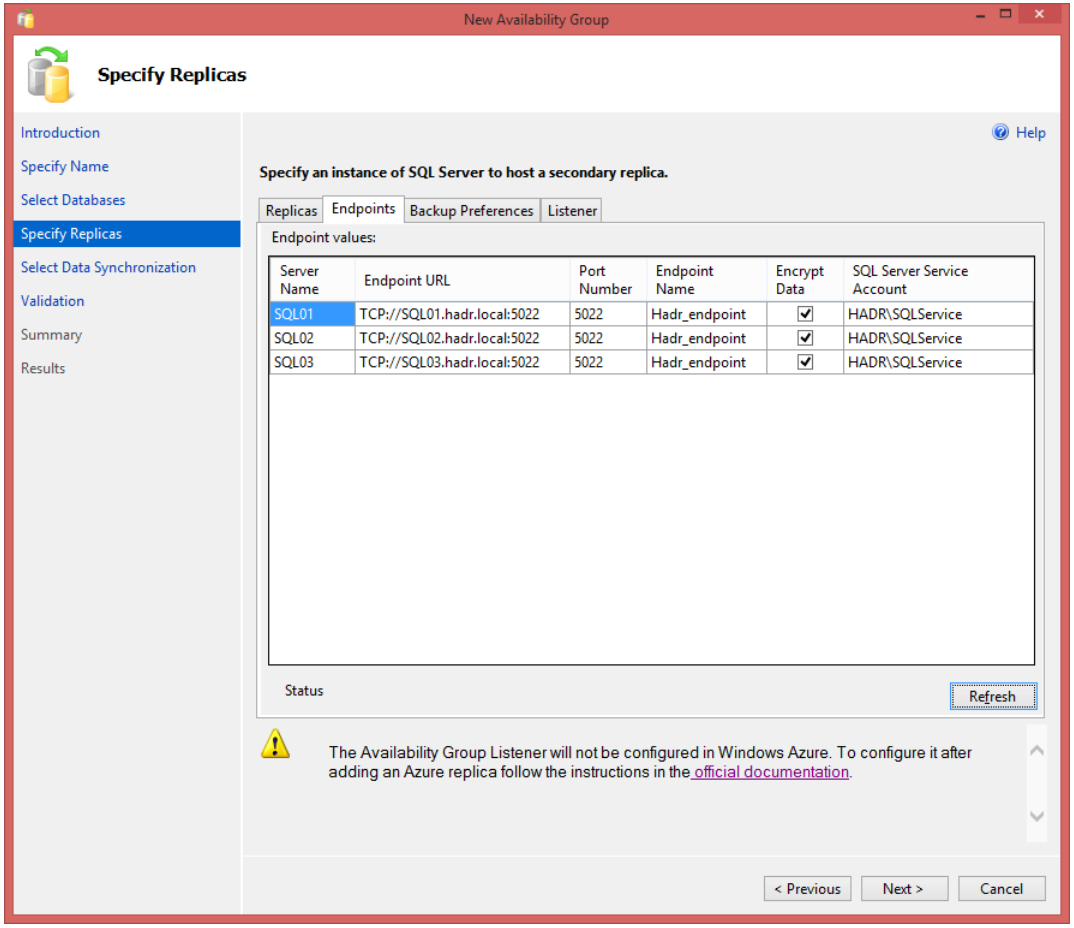
Readable secondary: No
In the secondary role, this availability replica will not allow any connections.

The Availability Group Listener will not be configured in Windows Azure. To configure it after adding an Azure replica follow the instructions in the official documentation.

< Previous | Next > | Cancel

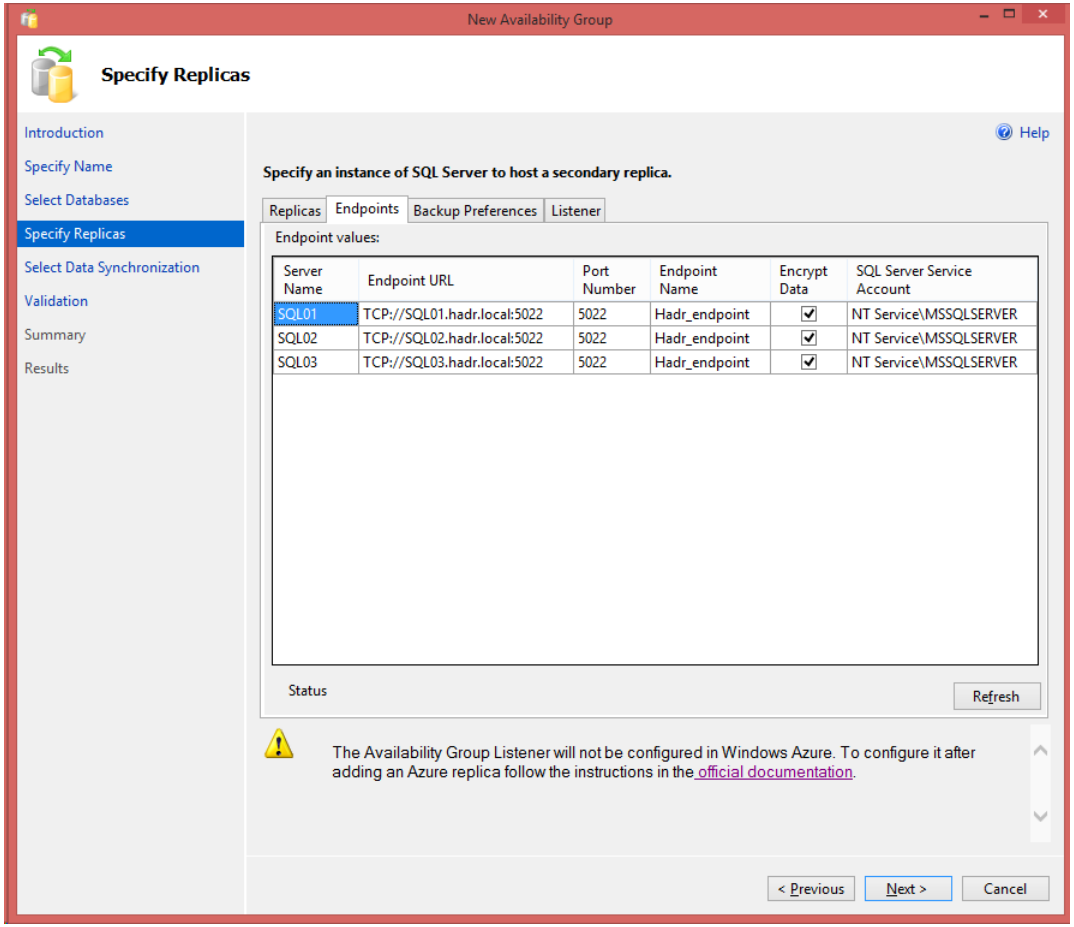
Step-by-step AG

▶ AG Wizard: configure Endpoints with Domain User service account



Step-by-step AG

▶ AG Wizard: configure Endpoints without Domain User service account



Step-by-step AG

▶ AG Wizard: configure Backup preferences

New Availability Group

Specify Replicas

Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Where should backups occur?

- **Prefer Secondary**
  Automated backups for this availability group should occur on a secondary replica. If there is no secondary replica available, backups will be performed on the primary replica.
- **Secondary only**
  All automated backups for this availability group must occur on a secondary replica.
- **Primary**
  All automated backups for this availability group must occur on the current primary replica.
- **Any Replica**
  Backups can occur on any replica in the availability group.

Replica backup priorities:

Server Instance	Backup Priority (Lowest=1, Highest=100)	Exclude Replica
SQL01	50	☐
SQL02	50	☐
SQL03	50	☐

The Availability Group Listener will not be configured in Windows Azure. To configure it after adding an Azure replica follow the instructions in the official documentation.

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: configure Listener

New Availability Group

Specify Replicas

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Specify your preference for an availability group listener that will provide a client connection point:

○ **Do not create an availability group listener now**

You can create the listener later using the Add Availability Group Listener dialog.

◉ **Create an availability group listener**

Specify your listener preferences for this availability group.

Listener DNS Name: SQLAG01

Port: 1433

Network Mode: Static IP

Subnet	IP Address
10.0.0.0/24	10.0.0.20
10.0.1.0/24	10.0.1.20

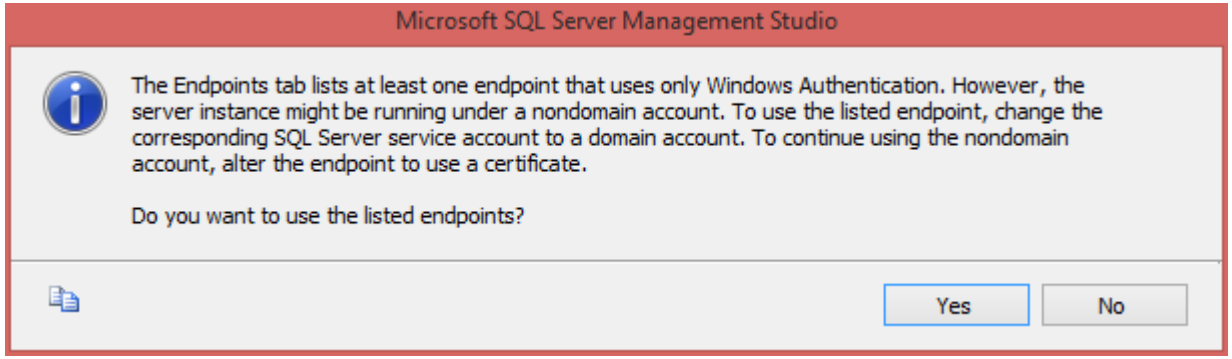
Add... | Remove

The Availability Group Listener will not be configured in Windows Azure. To configure it after adding an Azure replica follow the instructions in the official documentation.

< Previous | Next > | Cancel

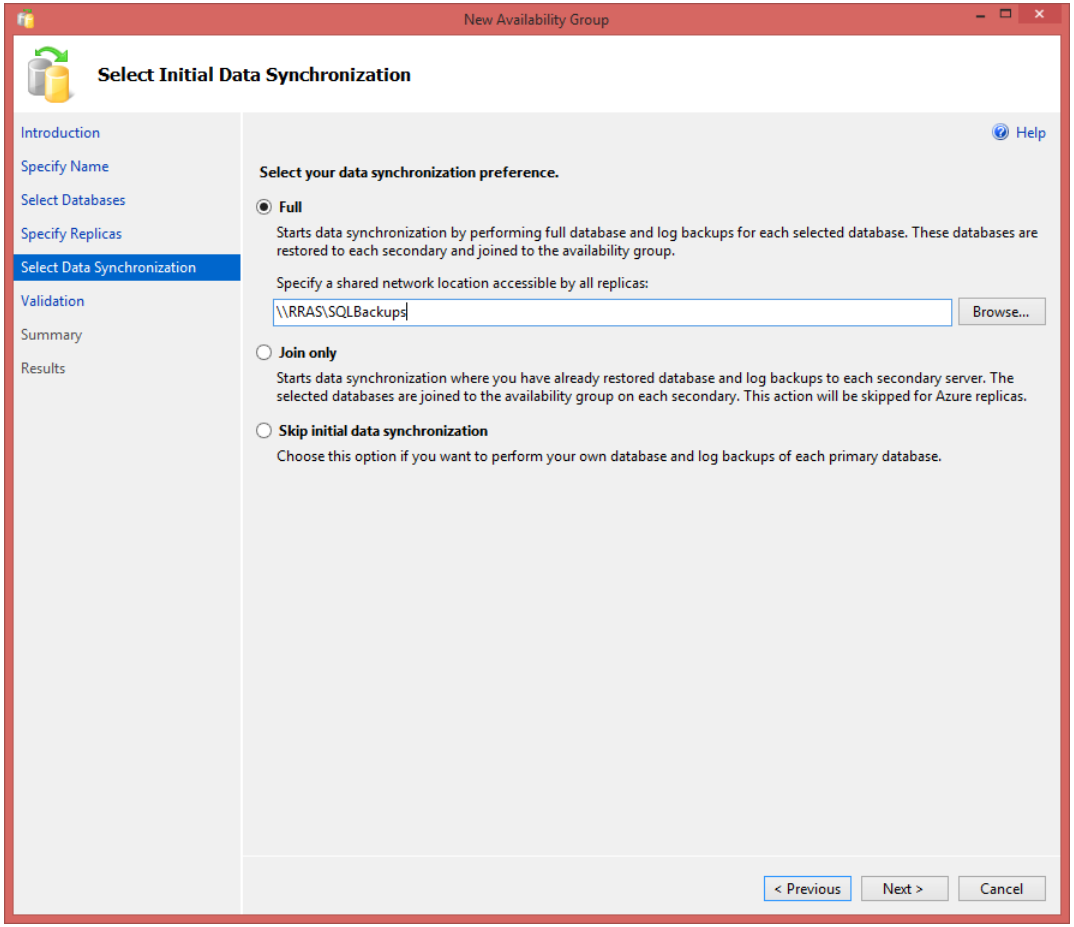
Step-by-step AG

▶ AG Wizard: warning about services accounts



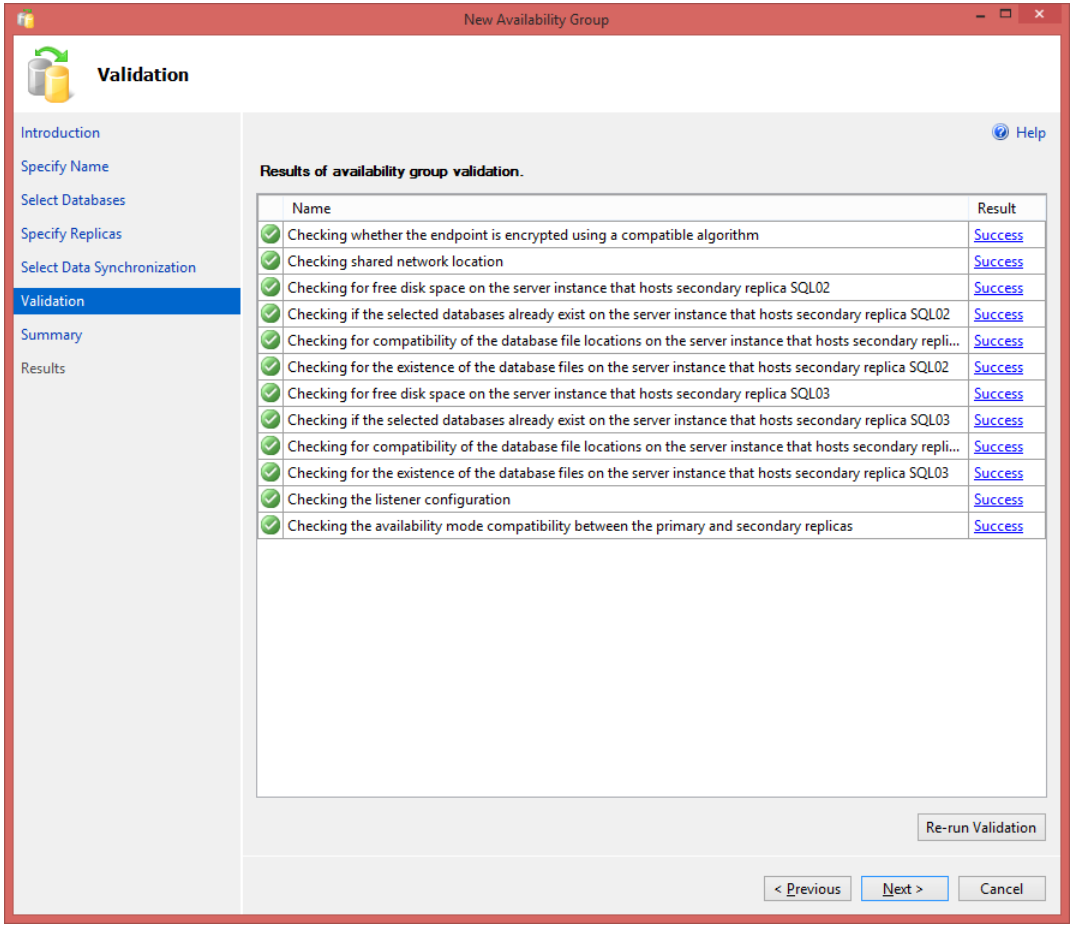
Step-by-step AG

▶ AG Wizard: configure data synchronization



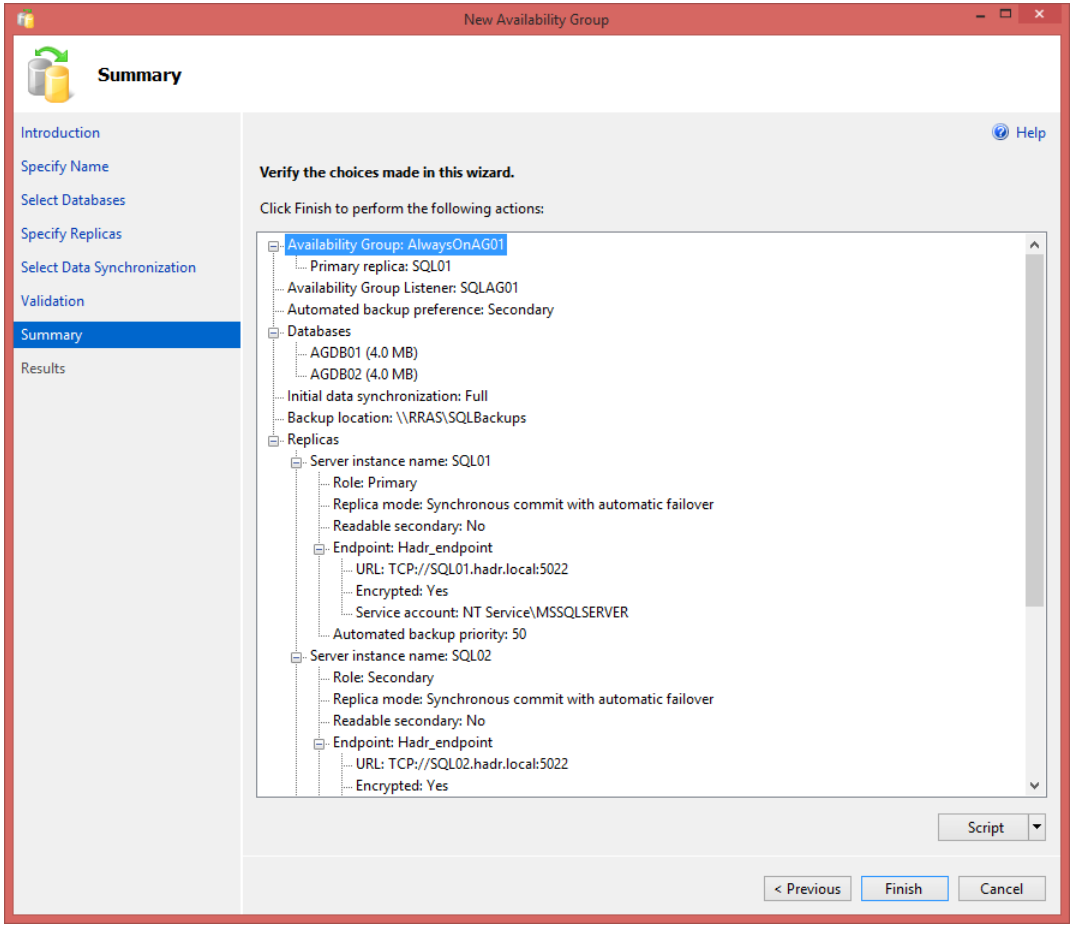
Step-by-step AG

▶ AG Wizard: Validation



Step-by-step AG

▶ AG Wizard: Summary



Step-by-step AG

▶ AG Wizard: Script

SQLQuery3.sql - SQL01.master (ghotz (52))* - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQL01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

Object Explorer | Registered Servers

SQLQuery3.sql - SQ...master (ghotz (52))*

```
--- YOU MUST EXECUTE THE FOLLOWING SCRIPT IN SQLCMD MODE.
:Connect SQL01
USE [master]
GO

CREATE LOGIN [HADR\SQLService] FROM WINDOWS
GO

:Connect SQL02
USE [master]
GO

CREATE LOGIN [HADR\SQLService] FROM WINDOWS
GO

:Connect SQL03
USE [master]
GO

CREATE LOGIN [HADR\SQLService] FROM WINDOWS
GO
```

100 %

Connected. (1/1) | SQL01 (12.0 SP1) | ghotz (52) | master | 00:00:00 | 0 rows

Ready | Ln 20 | Col 44 | Ch 44 | INS

Step-by-step AG

▶ AG Wizard: Progress

New Availability Group

Progress

Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results

Help

Configuring endpoints.

Name	Status
Creating logins on 'SQL01'.	Success
Creating logins on 'SQL02'.	Success
Creating logins on 'SQL03'.	Success
Configuring endpoints.	In Progress
Starting the 'AlwaysOn_health' XEvent session on 'SQL01'.	Not started
Starting the 'AlwaysOn_health' XEvent session on 'SQL02'.	Not started
Starting the 'AlwaysOn_health' XEvent session on 'SQL03'.	Not started
Creating availability group 'AlwaysOnAG01'.	Not started
Waiting for availability group 'AlwaysOnAG01' to come online.	Not started
Create Availability Group Listener 'SQLAG01'.	Not started
Joining secondary replicas to availability group 'AlwaysOnAG01'.	Not started
Validating WSFC quorum vote configuration.	Not started
Creating a full backup for 'AGDB01'.	Not started
Restoring 'AGDB01' on 'SQL02'.	Not started
Restoring 'AGDB01' on 'SQL03'.	Not started
Backing up log for 'AGDB01'.	Not started
Restoring 'AGDB01' log on 'SQL02'.	Not started
Joining 'AGDB01' to availability group 'AlwaysOnAG01' on 'SQL02'.	Not started
Restoring 'AGDB01' log on 'SQL03'.	Not started

Fewer details

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: Results

New Availability Group

Results

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

The wizard completed successfully.

Summary:

Name	Result
Creating logins on 'SQL01'.	Success
Creating logins on 'SQL02'.	Success
Creating logins on 'SQL03'.	Success
Configuring endpoints.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL01'.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL02'.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL03'.	Success
Creating availability group 'AlwaysOnAG01'.	Success
Waiting for availability group 'AlwaysOnAG01' to come online.	Success
Create Availability Group Listener 'SQLAG01'.	Success
Joining secondary replicas to availability group 'AlwaysOnAG01'.	Success
Validating WSFC quorum vote configuration.	Success
Creating a full backup for 'AGDB01'.	Success
Restoring 'AGDB01' on 'SQL02'.	Success
Restoring 'AGDB01' on 'SQL03'.	Success
Backing up log for 'AGDB01'.	Success
Restoring 'AGDB01' log on 'SQL02'.	Success
Joining 'AGDB01' to availability group 'AlwaysOnAG01' on 'SQL02'.	Success
Restoring 'AGDB01' log on 'SQL03'.	Success

< Previous | Next > | Close

Step-by-step AG

- AG open Dashboard

Solution1 - Microsoft SQL Server Management Studio

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

SQL01 (SQL Server 12.0.4100.1 - ghotz)
- Databases
  - System Databases
  - Database Snapshots
  - AGDB01
  - AGDB02
- Security
- Server Objects
- Replication
- AlwaysOn High Availability
  - Availability Groups
    - AlwaysOnAG01 (Primary
      - Availability Replicas
        - SQL01 (Primary)
        - SQL02 (Secondar
        - SQL03 (Secondar
      - Availability Database
        - AGDB01
        - AGDB02
      - Availability Group Li
        - SQLAG01
- Management
- Integration Services Catalogs
- SQL Server Agent

Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
Start PowerShell
Reports
Delete...
Refresh
Properties

Object Explorer | Registered Servers

Ready

Step-by-step AG

▶ AG Dashboard

AlwaysOnAG01:SQL01 - Microsoft SQL Server Management Studio

File Edit View Debug Tools Window Help

New Query

Object Explorer

Connect

- SQL01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Primary)
        - Availability Replicas
          - SQL01 (Primary)
          - SQL02 (Secondary)
          - SQL03 (Secondary)
        - Availability Databases
          - AGDB01
          - AGDB02
        - Availability Group Listeners
          - SQLAG01
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

Object Explorer | Registered Servers

AlwaysOnAG01:SQL01

AlwaysOnAG01: hosted by SQL01 (Replica role: Primary)

Last updated: 5/22/2015 7:48:25 PM

Auto refresh: on

Availability group state: Healthy
Primary instance: SQL01
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability replica: Add/Remove Columns

Name	Role	Failover Mode	Synchronization State	Issues
SQL01	Primary	Automatic	Synchronized	
SQL02	Secon...	Automatic	Synchronized	
SQL03	Secon...	Manual	Synchronizing	

Group by Add/Remove Columns

Name	Replica	Synchronization State	Failover Readi...	Issues
SQL01				
AGDB01	SQL01	Synchronized	No Data Loss	
AGDB02	SQL01	Synchronized	No Data Loss	
SQL02				
AGDB01	SQL02	Synchronized	No Data Loss	
AGDB02	SQL02	Synchronized	No Data Loss	
SQL03				
AGDB01	SQL03	Synchronizing	Data Loss	
AGDB02	SQL03	Synchronizing	Data Loss	

Ready

Step-by-step AG

- Cluster configuration of AG after installation

Failover Cluster Manager

File Action View Help

Failover Cluster Manager
- SQLCLUSTER.hadr.local
  - Roles
  - Nodes
  - Storage
  - Networks
  - Cluster Events

Roles (1)

Search | Queries

Name	Status	Type	Owner Node	Priority	Information
AlwaysOnAG01	Running	Other	SQL01	Medium	

AlwaysOnAG01 Preferred Owners: User Settings

Name	Status	Information
Other Resources		
AlwaysOnAG01	Online	
Server Name		
Name: SQLAG01	Online	
IP Address: 10.0.0.20	Online	
IP Address: 10.0.1.20	Offline	

Summary | Resources

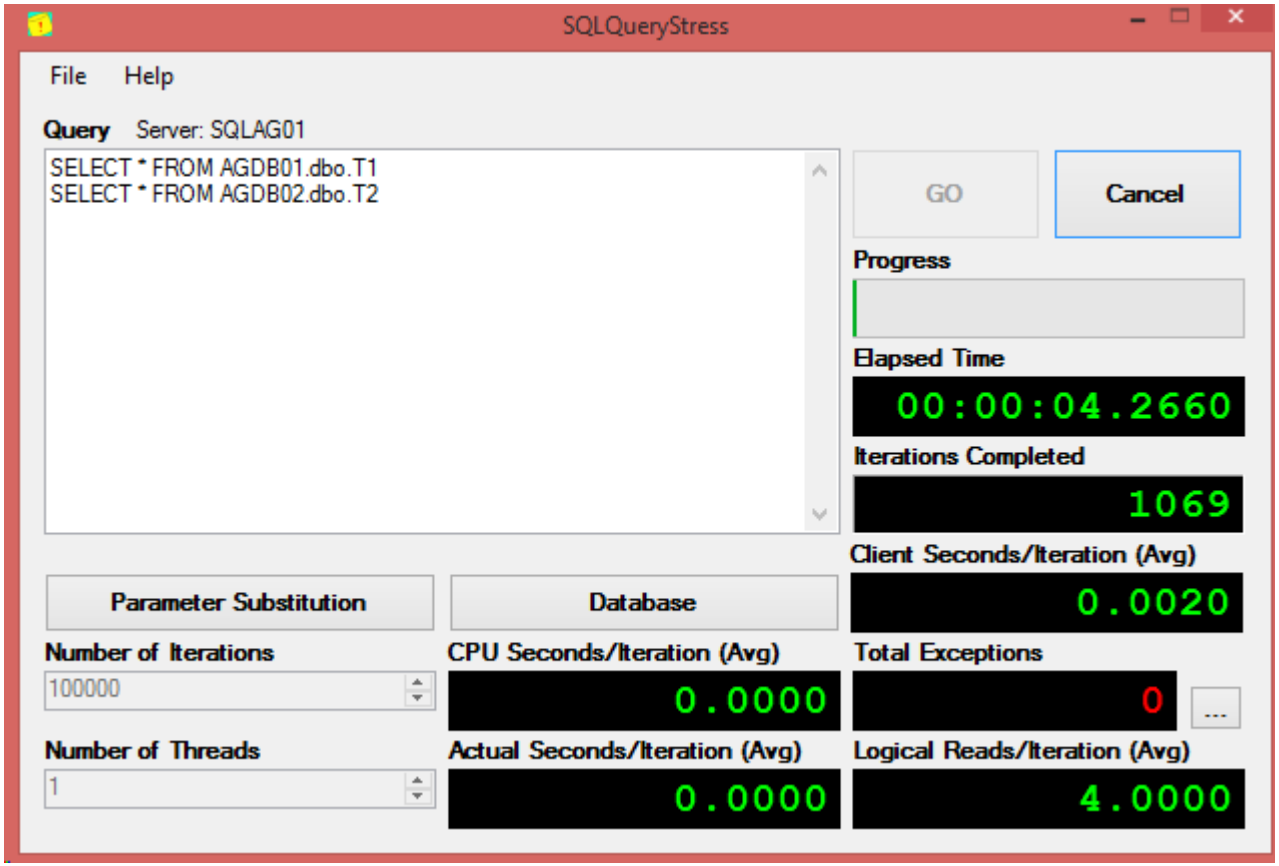
Actions

Roles
- Configure Role...
- Virtual Machines...
- Create Empty Role
- View
- Refresh
- Help

ITA 7:52 PM

Step-by-step AG

- Simulate a failure: start a workload



Step-by-step AG

- Simulate a failure: start a manual Failover

AlwaysOnAG01:SQL01 - Microsoft SQL Server Management Studio

File Edit View Debug Tools Window Help

New Query

Object Explorer

Connect

SQL01 (SQL Server 12.0.4100.1 - ghotz)
- Databases
  - System Databases
  - Database Snapshots
  - AGDB01 (Synchronized)
  - AGDB02 (Synchronized)
- Security
- Server Objects
- Replication
- AlwaysOn High Availability
  - Availability Groups
    - AlwaysOnAG01 (Primary)
      - Availability Replic
        - SQL01 (Primar
        - SQL02 (Secon
        - SQL03 (Secon
      - Availability Datab
        - AGDB01
        - AGDB02
      - Availability Group
        - SQLAG01
- Management
- Integration Services Catalogs
- SQL Server Agent

Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
Start PowerShell
Reports
Delete...
Refresh
Properties

Object Explorer | Registered Servers

AlwaysOnAG01:SQL01

AlwaysOnAG01: hosted by SQL01 (Replica role: Primary)

Last updated: 5/22/2015 11:37:53 PM

Auto refresh: on

Availability group state: Healthy
Primary instance: SQL01
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability replica: Add/Remove Columns

Name	Role	Failover Mode	Synchronization State	Issues
SQL01	Primary	Automatic	Synchronized	
SQL02	Secon...	Automatic	Synchronized	
	Secon...	Manual	Synchronizing	

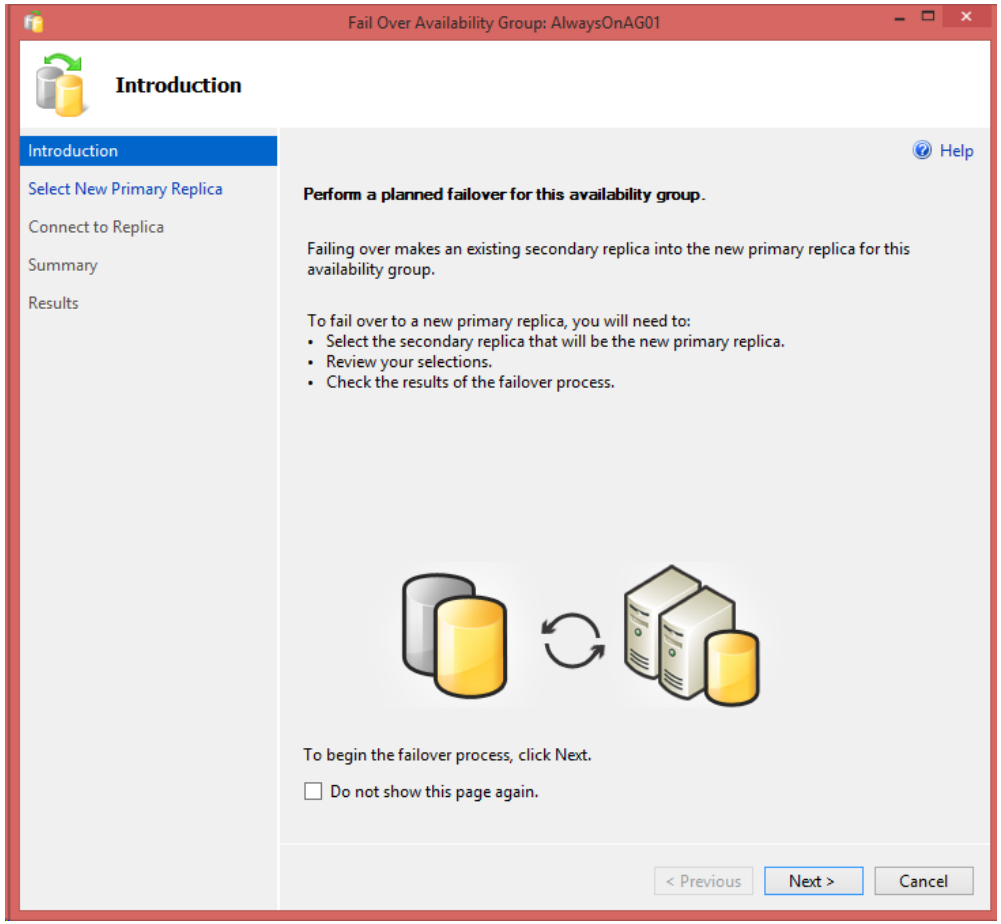
Add/Remove Columns

	Replica	Synchronization State	Failover Readi...	Issues
	SQL01	Synchronized	No Data Loss	
	SQL01	Synchronized	No Data Loss	
	SQL02	Synchronized	No Data Loss	
	SQL02	Synchronized	No Data Loss	
	SQL03	Synchronizing	Data Loss	
AGDB02	SQL03	Synchronizing	Data Loss	

Ready

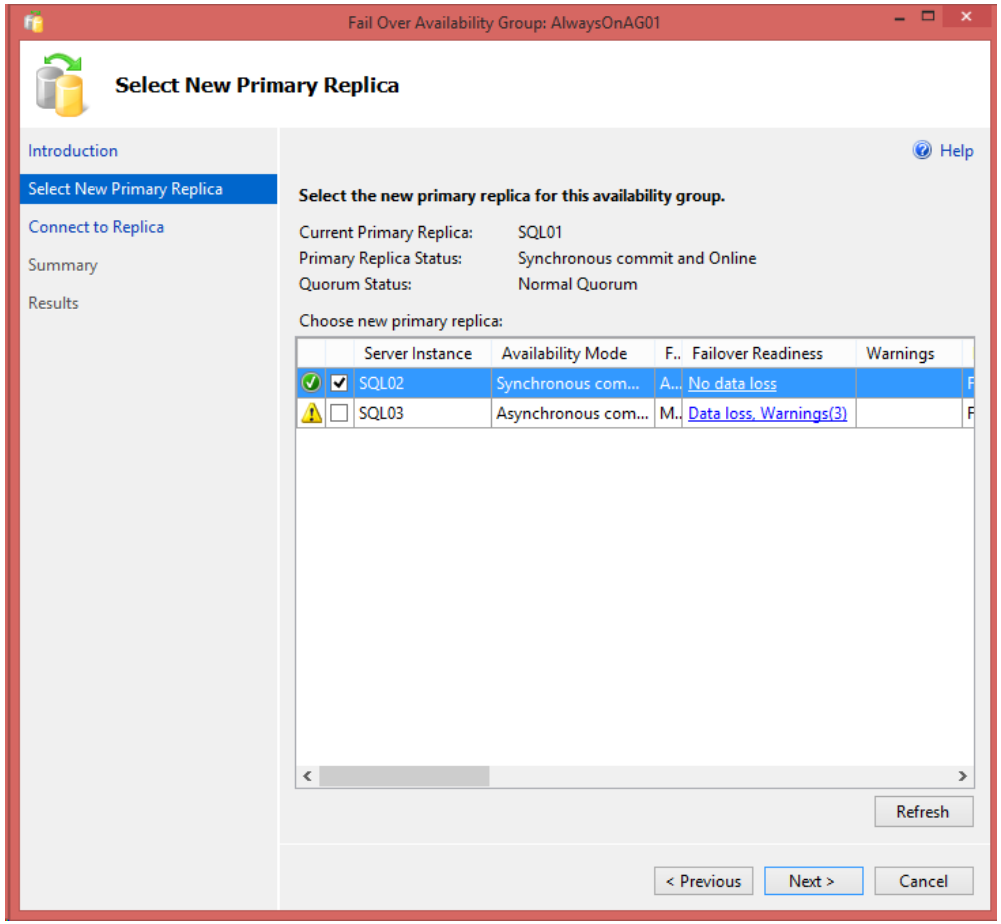
Step-by-step AG

▶ Failover Wizard: Introduction



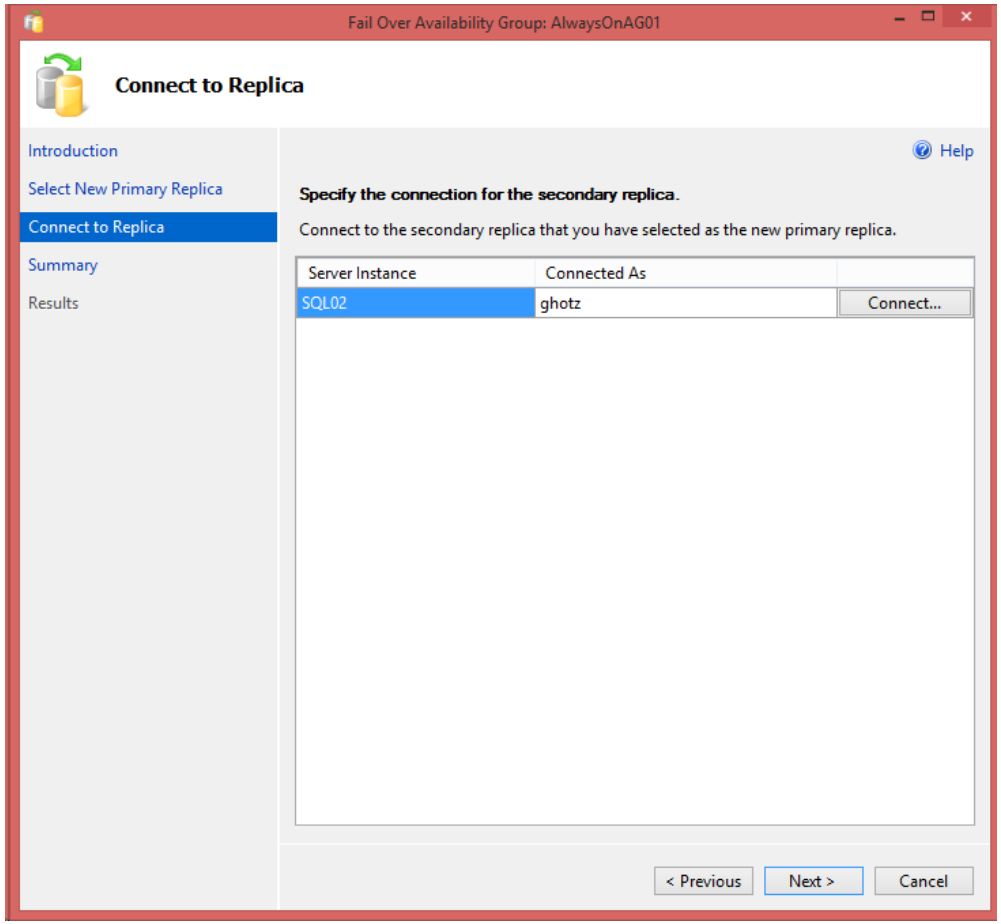
Step-by-step AG

▶ Failover Wizard: select new Primary Replica



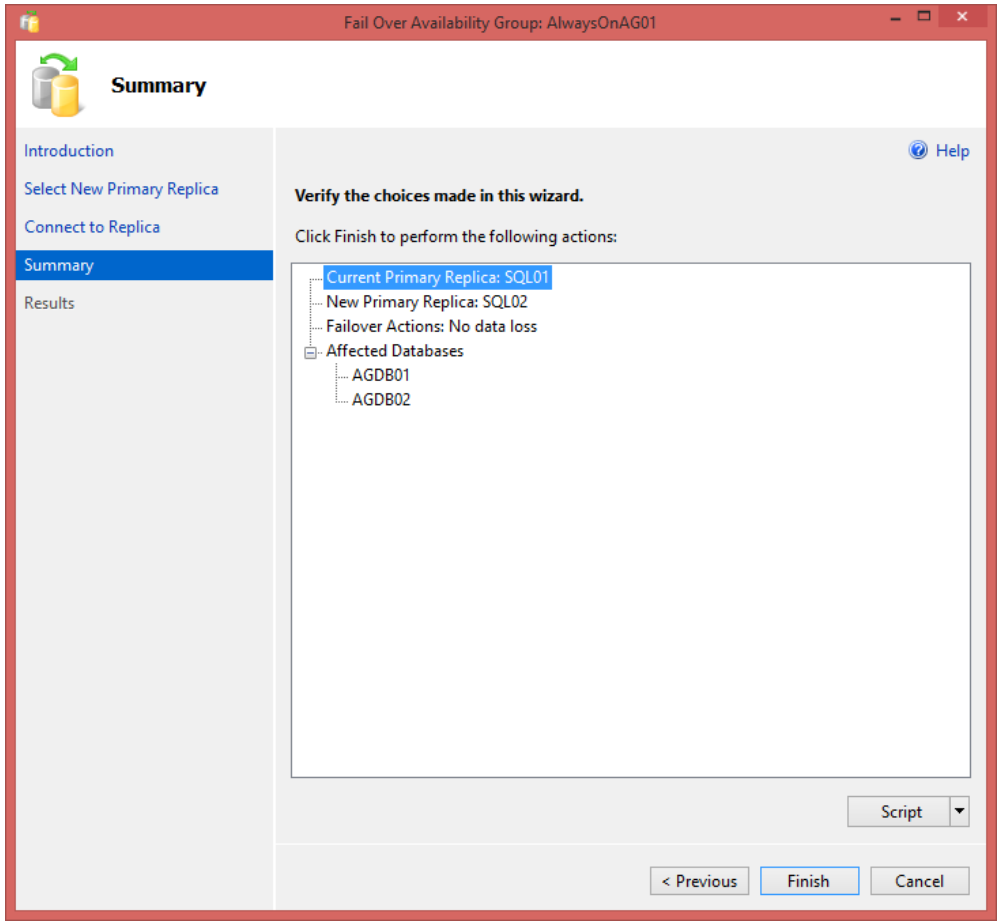
Step-by-step AG

▶ Failover Wizard: connect to Secondary Replica



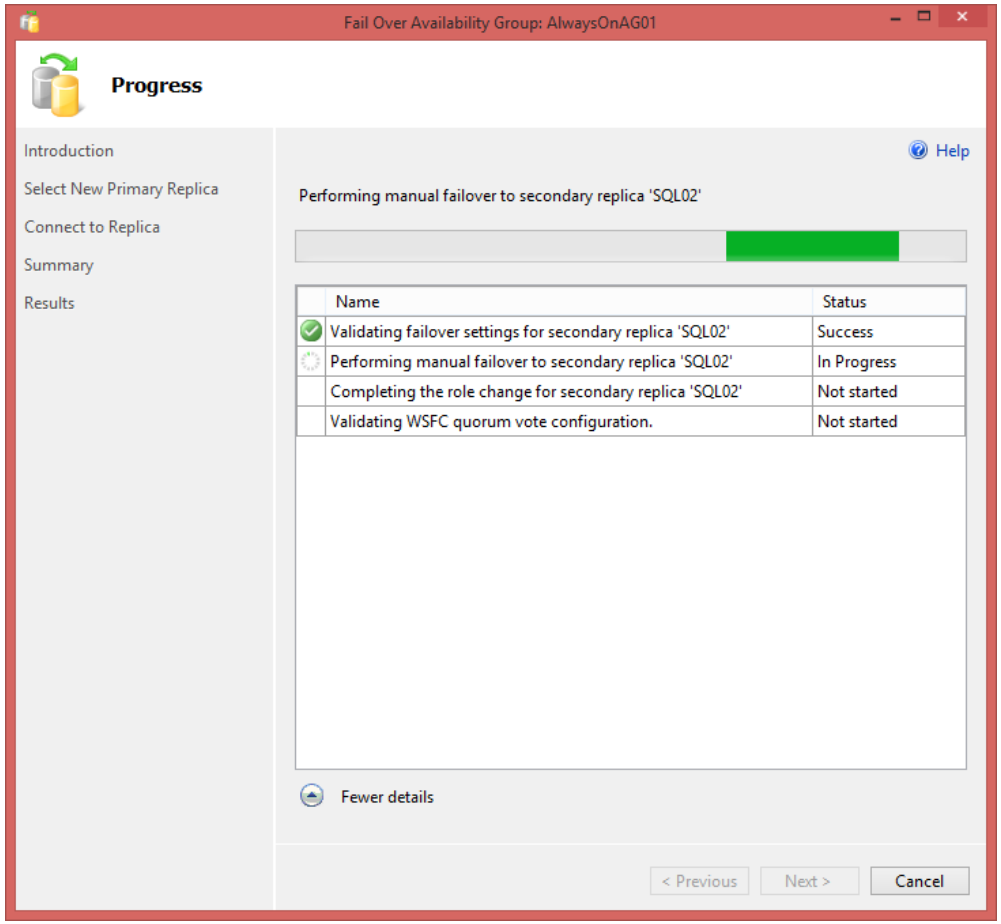
Step-by-step AG

▶ Failover Wizard: Summary



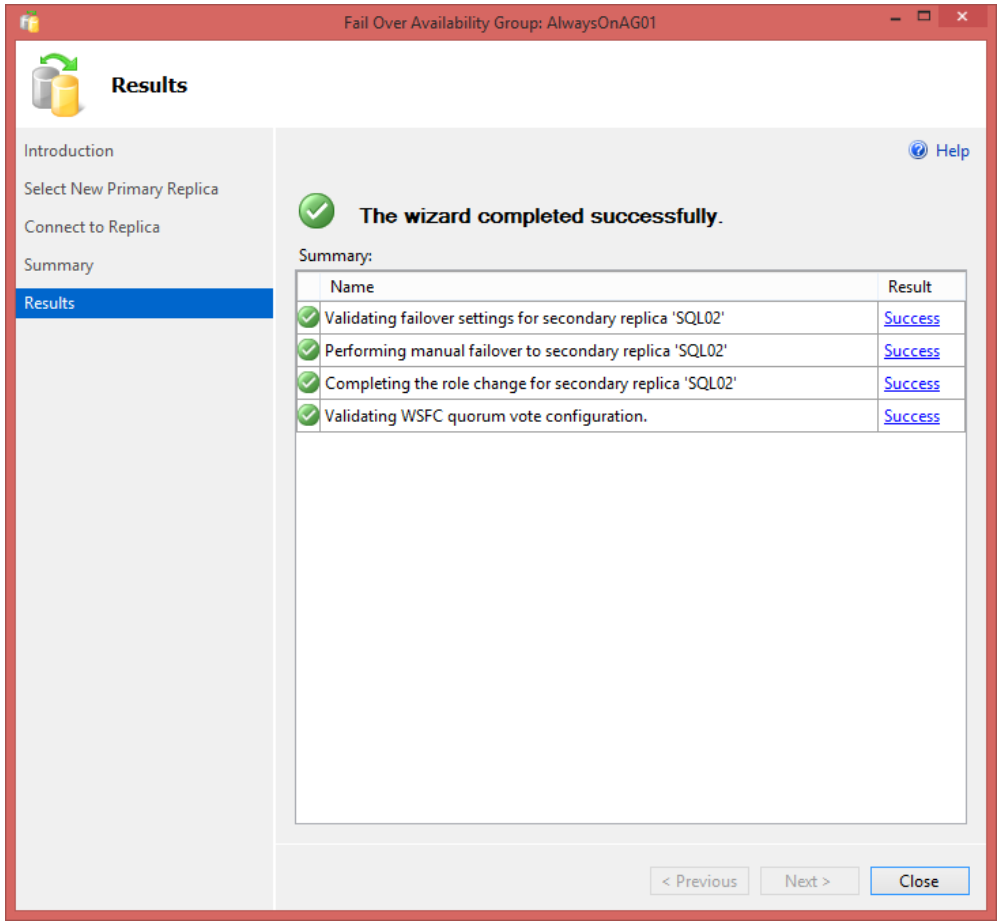
Step-by-step AG

▶ Failover Wizard: Progress



Step-by-step AG

▶ Failover Wizard: Complete



Step-by-step AG

- Simulate a failure: Dashboard on new Secondary Replica

AlwaysOnAG01:SQL01 - Microsoft SQL Server Management Studio

File Edit View Debug Tools Window Help

New Query

Object Explorer

Connect

SQL01 (SQL Server 12.0.4100.1 - ghotz)
- Databases
  - System Databases
  - Database Snapshots
  - AGDB01 (Synchronized)
  - AGDB02 (Synchronized)
- Security
- Server Objects
- Replication
- AlwaysOn High Availability
  - Availability Groups
    - AlwaysOnAG01 (Secondary)
      - Availability Replicas
      - Availability Databases
      - Availability Group Listeners
- Management
- Integration Services Catalogs
- SQL Server Agent

Object Explorer | Registered Servers

AlwaysOnAG01:SQL01

AlwaysOnAG01: hosted by SQL01 (Replica role: Secondary)

Last updated: 5/22/2015 11:41:52 PM

Auto refresh: on

Availability group state: No data available on secondary replicas.

Primary instance: SQL02

Failover mode: Automatic

Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard

View AlwaysOn Health Events

View Cluster Quorum Information

Availability replica: Add/Remove Columns

Name	Role	Failover Mode	Synchronization State	Issues
SQL01	Secon...	Automatic	Synchronized	

Group by

Add/Remove Columns

Name	Replica	Synchronization State	Failover Readi...	Issues
SQL01				
AGDB01	SQL01	Synchronized	No Data Loss	
AGDB02	SQL01	Synchronized	No Data Loss	

Ready

Step-by-step AG

- Simulate a failure: Dashboard on new Primary Replica

AlwaysOnAG01:SQL02 - Microsoft SQL Server Management Studio

File Edit View Tools Window Help

New Query

Object Explorer

Connect

- SQL01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Secondary)
        - Availability Replicas
        - Availability Databases
        - Availability Group Listeners
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

Object Explorer | Registered Servers

AlwaysOnAG01:SQL02 | AlwaysOnAG01:SQL01

AlwaysOnAG01: hosted by SQL02 (Replica role: Primary)

Last updated: 5/22/2015 11:44:02 PM

Auto refresh: on

Availability group state: Healthy
Primary instance: SQL02
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability replica: Add/Remove Columns

Name	Role	Failover Mode	Synchronization State	Issues
SQL01	Secon...	Automatic	Synchronized	
SQL02	Primary	Automatic	Synchronized	
SQL03	Secon...	Manual	Synchronizing	

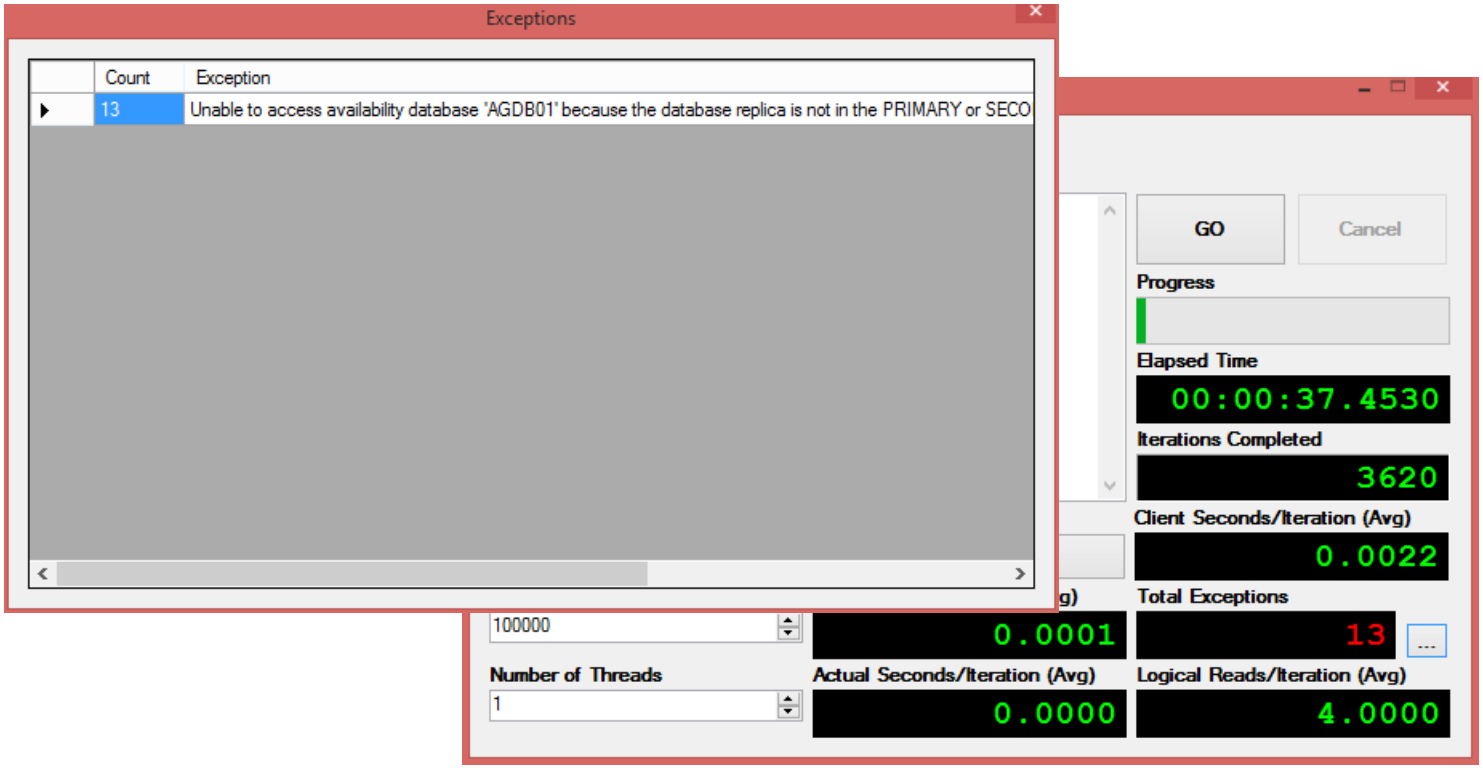
Group by Add/Remove Columns

Name	Replica	Synchronization State	Failover Readi...	Issues
SQL01				
AGDB01	SQL01	Synchronized	No Data Loss	
AGDB02	SQL01	Synchronized	No Data Loss	
SQL02				
AGDB01	SQL02	Synchronized	No Data Loss	
AGDB02	SQL02	Synchronized	No Data Loss	
SQL03				
AGDB01	SQL03	Synchronizing	Data Loss	
AGDB02	SQL03	Synchronizing	Data Loss	

Ready

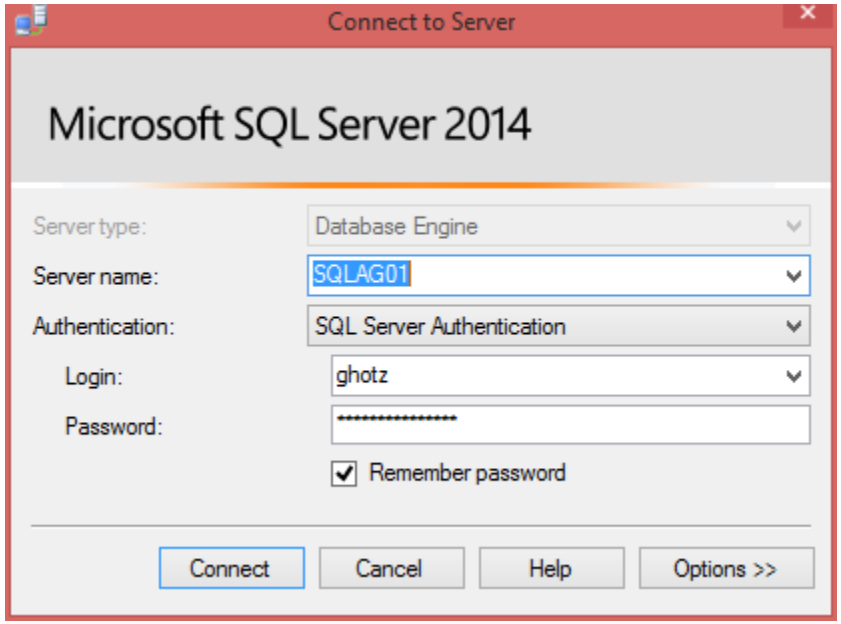
Step-by-step AG

- Simulate a failure: clients exceptions



Step-by-step AG

▶ Accessing AG: connecting to the Listener's VNN



Step-by-step AG

- Accessing AG: querying the Listener's VNN

ag.04.test.01.simple-queries.sql - SQLAG01.master (ghotz (62)) - Microsoft SQL Server Management Studio

File Edit View Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

ag.04.test.01.simple....master (ghotz (62))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL02

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

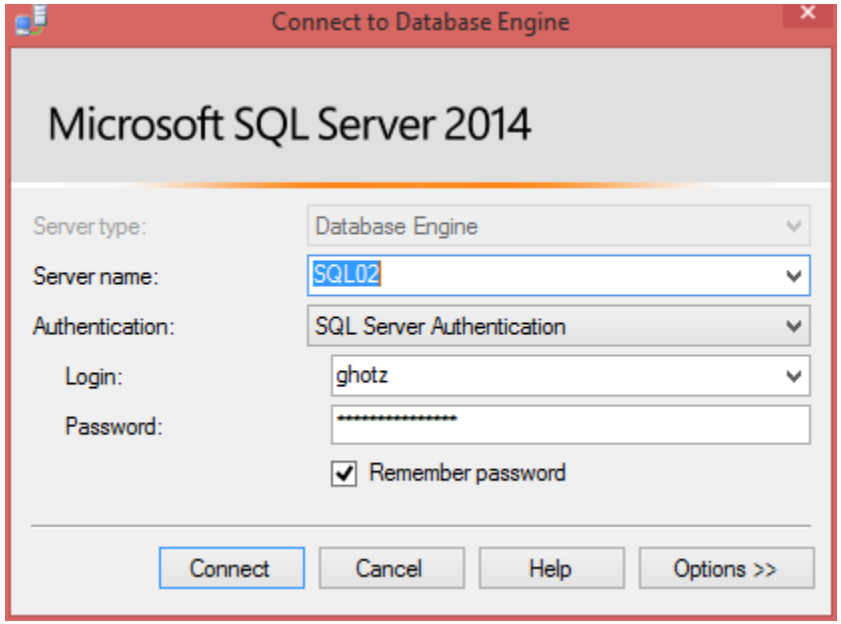
Object Explorer | Registered Servers

Query executed successfully. | SQLAG01 (12.0 SP1) | ghotz (62) | master | 00:00:00 | 11 rows

Ready

Step-by-step AG

▶ Accessing AG: connecting directly to Secondary Replica



Step-by-step AG

▶ Accessing AG: Secondary Replica not accessible

ag.04.test.01.simple-queries.sql - SQL02.master (ghotz (62)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

ag.04.test.01.simple....master (ghotz (62))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL02

(1 row(s) affected)

Msg 976, Level 14, State 1, Line 3
The target database, 'AGDB01', is participating in an availability group and is currently not accessible
for queries. Either data movement is suspended or the availability replica is not enabled for read access.
To allow read-only access to this and other databases in the availability group, enable read access to one
or more secondary availability replicas in the group.  For more information, see the ALTER AVAILABILITY GROUP
statement in SQL Server Books Online.
```

100 %

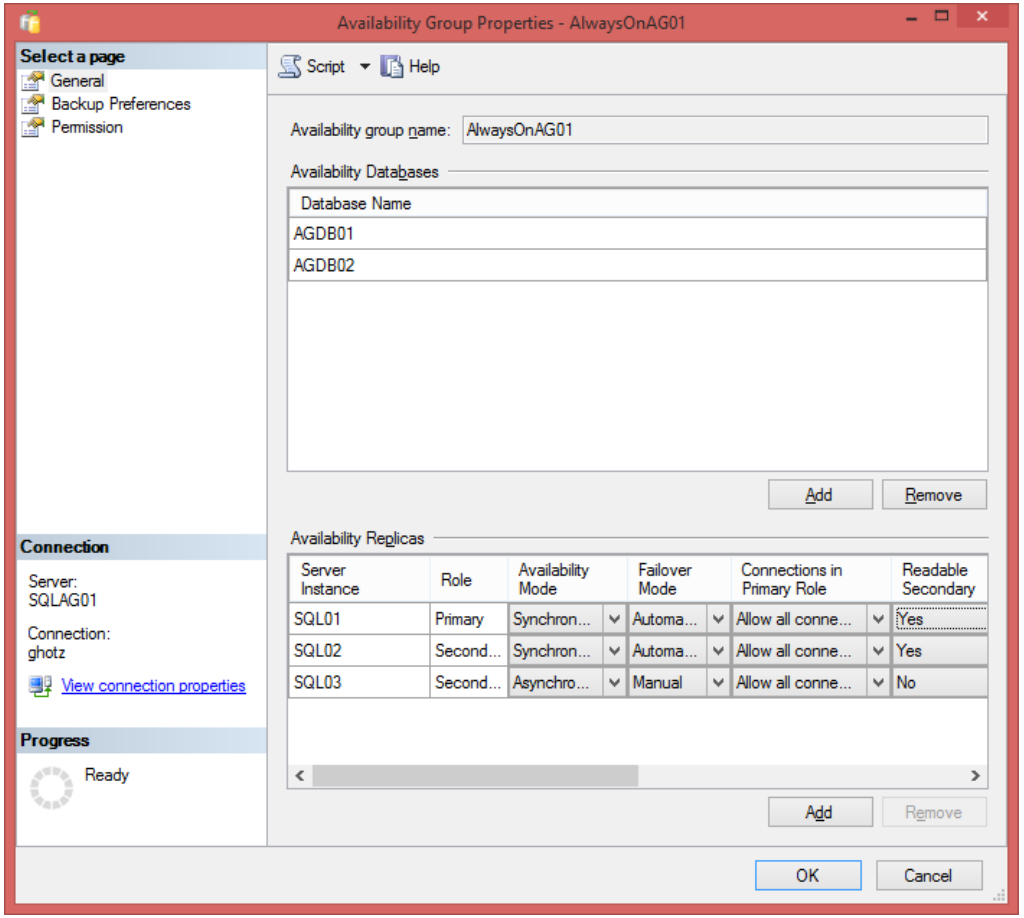
Object Explorer | Registered Servers

Query completed with errors. | SQL02 (12.0 SP1) | ghotz (62) | master | 00:00:00 | 1 rows

Ready | Ln 12 | Col 38 | Ch 38 | INS

Step-by-step AG

▶ Accessing AG: configure readable Secondary Replica



Step-by-step AG

▶ Accessing AG: readable Secondary Replica

ag.04.test.01.simple-queries.sql - SQL02.master (ghotz (62)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

Object Explorer | Registered Servers

ag.04.test.01.simple....master (ghotz (62))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL02

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Query executed successfully. | SQL02 (12.0 SP1) | ghotz (62) | master | 00:00:00 | 11 rows

Ready | Ln 4 | Col 29 | Ch 29 | INS

Step-by-step AG

▶ Accessing AG: configure Application Intent

Availability Group Properties - AlwaysOnAG01

Select a page
- General
- Backup Preferences
- Permission

Script ▾ Help

Availability group name: AlwaysOnAG01

Availability Databases

Database Name
AGDB01
AGDB02

Add Remove

Availability Replicas

Server Instance	Role	Availabilit Mode	Failover Mode	Connections in Primary Role	Readable Secor
SQL01	Primary	Sy...	Automatic	Allow read/write connections	Read-intent only
SQL02	Secon...	Sy...	Automatic	Allow read/write connections	Read-intent only
SQL03	Secon...	As...	Manual	Allow all connections	No

Add Remove

Connection

Server:
SQLAG01

Connection:
ghotz

View connection properties

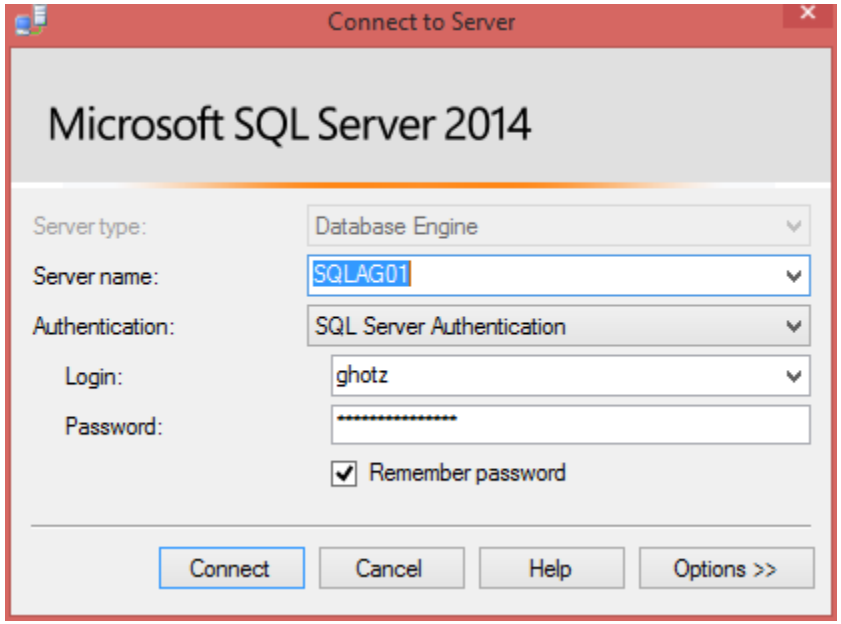
Progress

Ready

OK Cancel

Step-by-step AG

▶ Accessing AG: VNN without Application Intent



Step-by-step AG

- Accessing AG: VNN without Application Intent defaults to read/write

ag.04.test.01.simple-queries.sql - SQLAG01.master (ghotz (53)) - Microsoft SQL Server Management Studio

File Edit View Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

Object Explorer | Registered Servers

ag.04.test.01.simple....master (ghotz (53))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL01

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

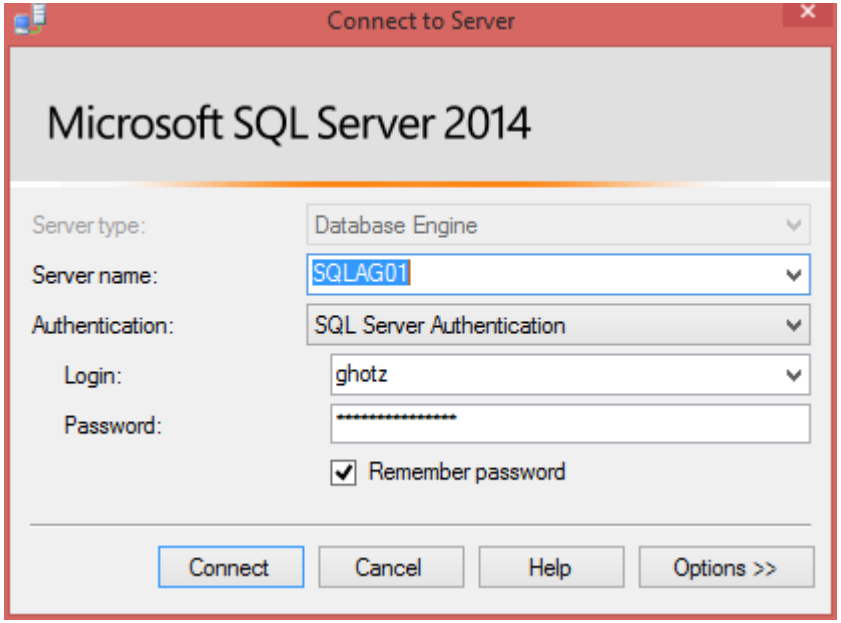
100 %

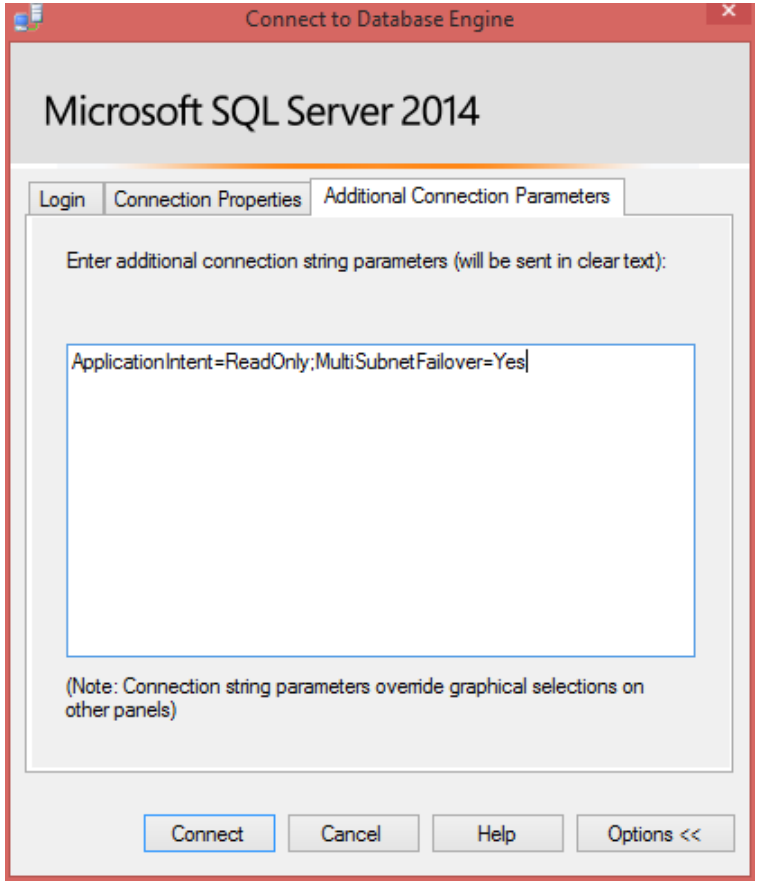
Query executed successfully. | SQLAG01 (12.0 SP1) | ghotz (53) | master | 00:00:00 | 11 rows

Ready

Step-by-step AG

▶ Accessing AG: VNN with Read Only Application Intent





Step-by-step AG

- Accessing AG: VNN with Read Only Application Intent still on Primary

ag.04.test.01.simple-queries.sql - SQLAG01.master (ghotz (53)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

ag.04.test.01.simple....master (ghotz (53))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL01

(1 row(s) affected)

Msg 979, Level 14, State 1, Line 3
The target database ('AGDB01') is in an availability group and currently does not allow read only connections.
For more information about application intent, see SQL Server Books Online.
```

100 %

Object Explorer | Registered Servers

Query completed with errors. | SQLAG01 (12.0 SP1) | ghotz (53) | master | 00:00:00 | 1 rows

Ready | Ln 9 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ Accessing AG: Read Only Routing configuration

ag.04.test.03.configure-read-only-routing.sql - SQLAG01.AGDB01 (ghotz (53)) - Microsoft SQL Server Management Studio

```
-- Configure Read Only Routing URLs
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL01' WITH (SECONDARY_ROLE(READ_ONLY_ROUTING_URL=N'TCP://sql01.hadr.local:1433'))
GO
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL02' WITH (SECONDARY_ROLE(READ_ONLY_ROUTING_URL=N'TCP://sql02.hadr.local:1433'))
GO

-- Configure Read Only Routing lists
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL01' WITH (PRIMARY_ROLE(READ_ONLY_ROUTING_LIST=(N'SQL02', N'SQL01')))
GO
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL02' WITH (PRIMARY_ROLE(READ_ONLY_ROUTING_LIST=(N'SQL01', N'SQL02')))
GO
```

```
Command(s) completed successfully.
```

Step-by-step AG

- Accessing AG: VNN with Read Only Application Intent again on Primary

ag.04.test.01.simple-queries.sql - SQLAG01.master (ghotz (62)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

ag.04.test.01.simple....master (ghotz (62))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL01

(1 row(s) affected)

Msg 979, Level 14, State 1, Line 3
The target database ('AGDB01') is in an availability group and currently does not allow read only connections.
For more information about application intent, see SQL Server Books Online.
```

100 %

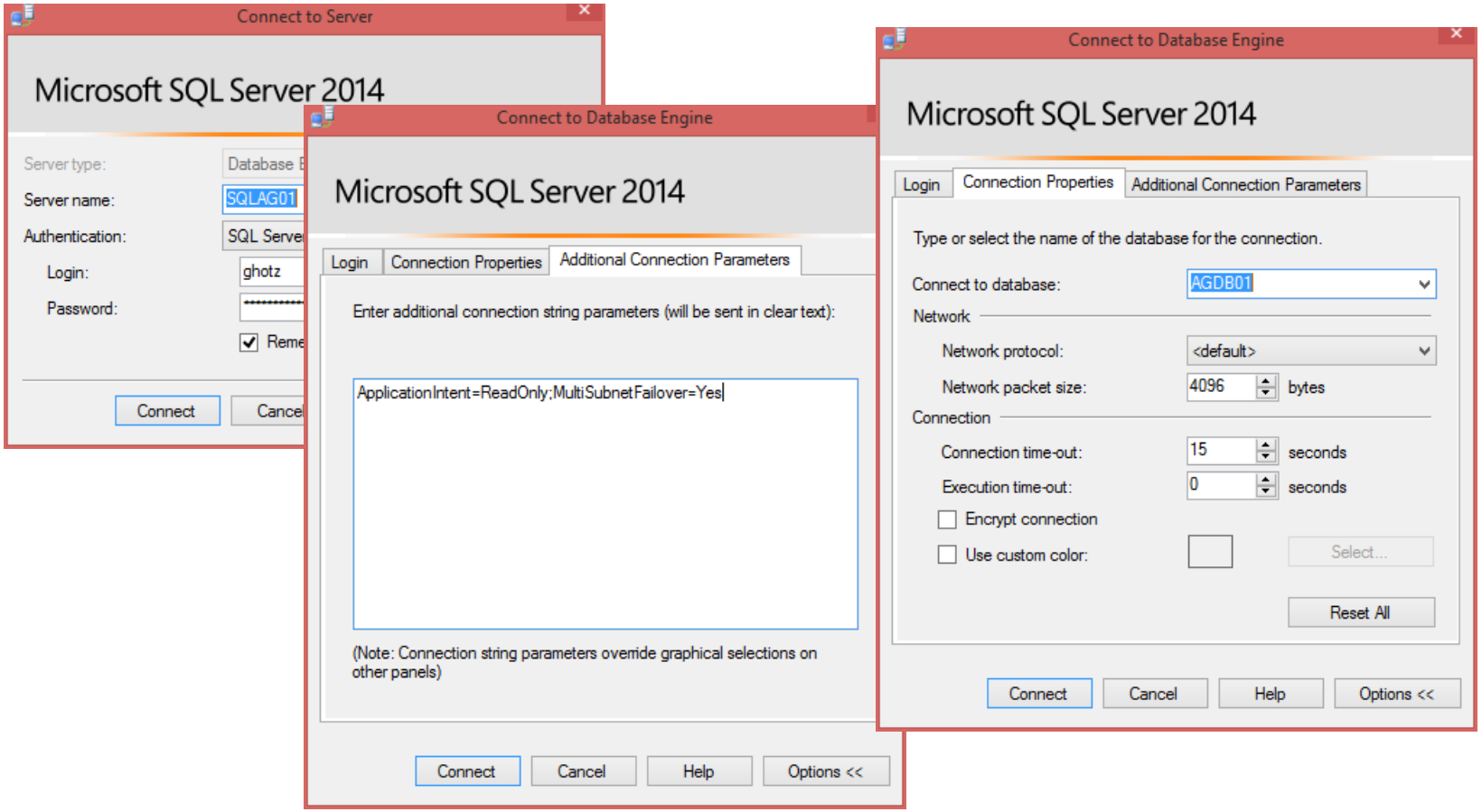
Object Explorer | Registered Servers

Query completed with errors. | SQLAG01 (12.0 SP1) | ghotz (62) | master | 00:00:00 | 1 rows

Ready | Ln 9 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ Accessing AG: VNN with Read Only Application Intent and database name



Step-by-step AG

- Accessing AG: Read Only Routing working!

ag.04.test.01.simple-queries.sql - SQLAG01.AGDB01 (ghotz (57)) - Microsoft SQL Server Management Studio

File Edit View Query Project Debug Tools Window Help

New Query | AGDB01 | Execute | Debug

Object Explorer

Connect

- SQLAG01 (SQL Server 12.0.4100.1 - ghotz)
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

ag.04.test.01.simpl...AGDB01 (ghotz (57))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL02

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Object Explorer | Registered Servers

Query executed successfully. | SQLAG01 (12.0 SP1) | ghotz (57) | AGDB01 | 00:00:00 | 11 rows

Ready | Ln 4 | Col 21 | Ch 21 | INS

AlwaysOn

▶ WSFC: Quorum, votes and DR site

- Quorum
  - Majority Node, Majority Node and Fileshare
  - Can’t use Quorum Disk without shared storage
- Windows 2008 e 2008 R2 + QFE
  - Allow to assign voting rights to nodes
  - Avoid scenarios where DR site takes over
  - Common scenario is to always failover manually to DR site i.e. DR site is not considered for HA

AlwaysOn

▶ WSFC: Asymmetric Storage

- Windows 2008 + QFE or 2008 R2 + SP1
- Disks shared only among some nodes
  - E.g. one SAN on primary site and one SAN on DR site, volumes shared only among local nodes while the cluster stretches across sites
- Allows new Quorum models
  - Asymmetric Disk Majority
  - Node Majority + Asymmetric Disk Majority

AlwaysOn

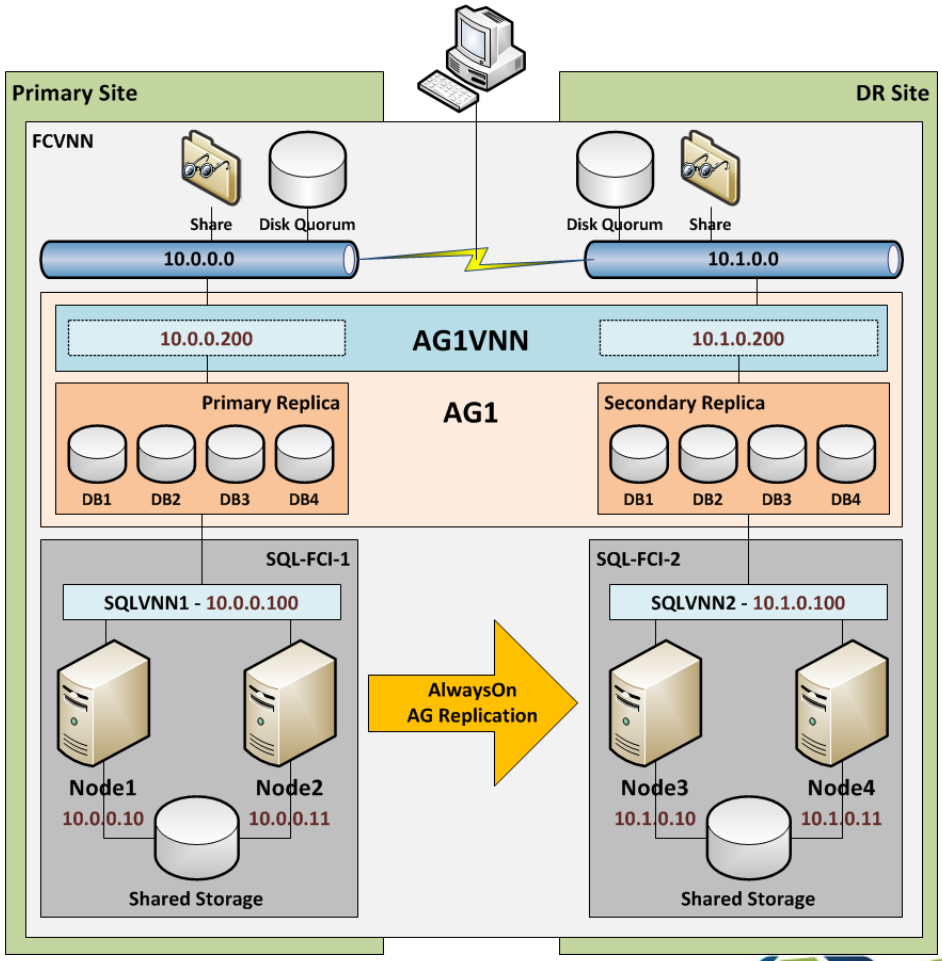
▶ Topology Example 2

Shared storage, regional and geo secondaries

Asynchronous data movement

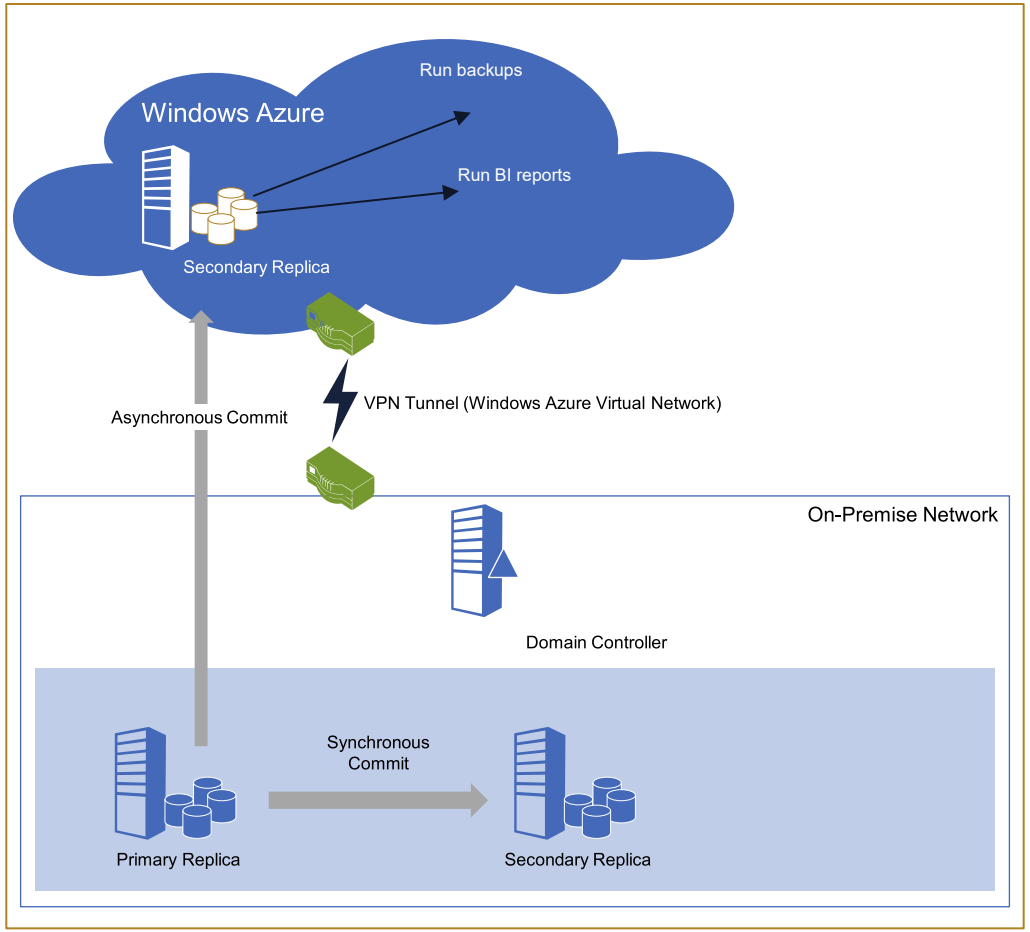
AlwaysOn

- Multi-site FCI+AG scenario



AlwaysOn AG

▶ Hybrid Cloud



AlwaysOn

▶ SQL Server 2012 Design Patterns

AlwaysOn Solution	Characteristics	Pre AlwaysOn corresponding solution
Multi-site FCI for HA and DR	• Shared Storage solution • Instance Level HA • Instance Level DR • Doesn't require database to be in FULL recovery model	Multi-site FCI using stretch VLAN
Availability Group for HA and DR	• Non-Shared Storage solution • (Group of) Database Level HA • (Group of) Database Level DR • DR replica can be Active Secondary • Requires database to be in FULL recovery model	Database Mirroring for Local HA and Log Shipping for DR
Failover Cluster Instance for local HA + Availability Group for DR	• Combined Shared Storage and Non-Shared Storage • Instance Level HA • (Group of) Database Level DR • DR replica can be Active Secondary • Requires database to be in FULL recovery model	FCI for Local HA and Database Mirroring for DR

AlwaysOn

▶ Availability Groups in SQL Server 2014

- Up to 8 secondary replicas (was 4)
  - still only 2 can be in synchronous mode
- Secondary readable replicas remain online
  - when disconnected from primary
  - When there's a quorum loss
- New DMV
  - sys.fn_hadr_is_primary_replica
- Add Azure Replica Wizard

AlwaysOn

▶ Availability Groups in SQL Server 2016

- Up to 3 synchronous secondary replicas (was 2)
- DTC Support
- Round robin balancing of secondary replicas

AlwaysOn

▶ Minimum HA/DR support for non Enterprise editions

- Server Core
- Log Shipping (except on Express)
- Database Mirroring
  - Deprecated!
  - Synchronous only for BI/Standard
  - Witness only for Web/Express
- Failover Clustering Instance
  - 2 nodes for BI/Standard
- Purely passive nodes licensing as in 2008

AlwaysOn

▶ Other Hybrid Cloud Scenarios

- Database Mirroring
  - VPN not needed for domain authentication
  - Can use certificate based authentication

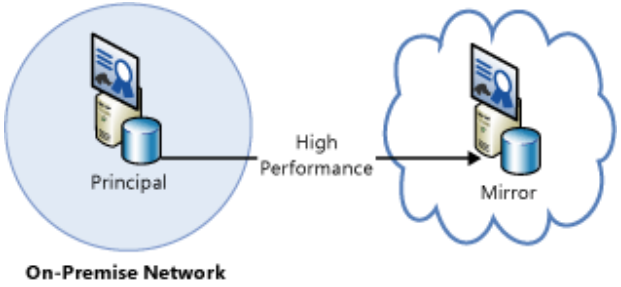


Image source: http://msdn.microsoft.com/en-us/library/jj870962.aspx

- Log Shipping

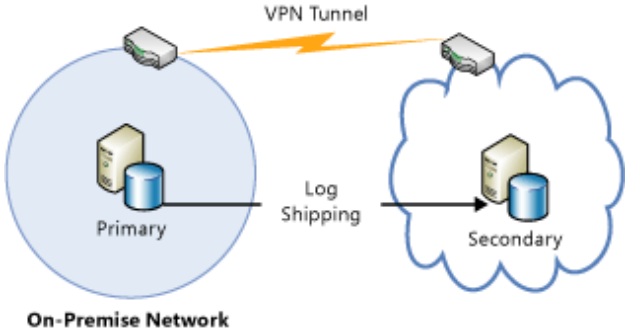


Image source: http://msdn.microsoft.com/en-us/library/jj870962.aspx

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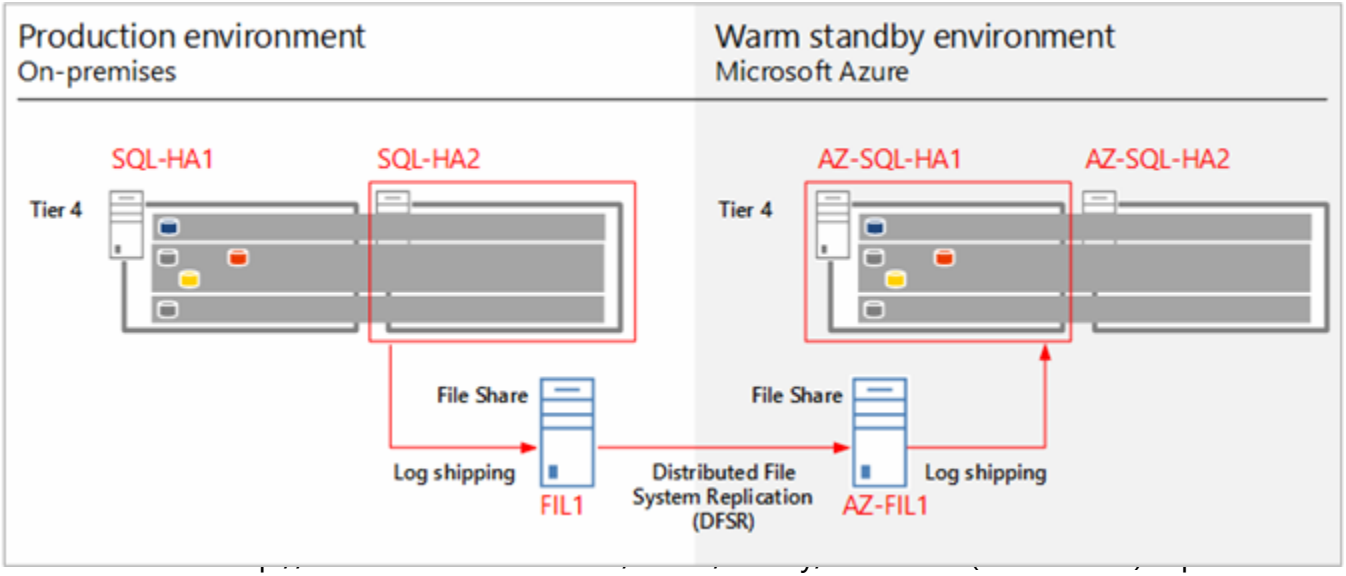
▶ AG+Log Shipping Scenario



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▶ Change in licensing

- Software Assurance needed for passive servers
  - "Beginning with SQL Server 2014, each active server licensed with SA coverage allows the installation of a single passive server used for fail-over support."
  - "The active server license (s) must be covered with SA, and allow for one passive secondary SQL Server, with up to the same amount of compute as the licensed active server, only."
  - Impacts everything: log shipping, mirroring, AG
  - From SQL Server 2014 Licensing Datasheet
    - http://download.microsoft.com/download/6/6/F/66FF3259-1466-4BBA-A505-2E3DA5B2B1FA/SQL_Server_2014_Licensing_Datasheet.pdf

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▶ AG Main Resources

- Books Online
- SQL AlwaysOn Blog
  - http://blogs.msdn.com/b/sqlalwayson
- CSS SQL Server Engineers
  - http://blogs.msdn.com/b/psssql/archive/tags/alwayson
- SQL CAT
  - http://sqlcat.com
- Previously named HADRON
  - Keep in mind when searching articles
  - Still present in some DMVs (_hadr_)

AlwaysOn

▶ AG Whitepapers series: AlwaysOn HA/DR architectures and design patterns

- AlwaysOn Architecture Guide: Building a High Availability and Disaster Recovery Solution by Using AlwaysOn Availability Groups
  - http://msdn.microsoft.com/en-us/library/jj191711.aspx
- AlwaysOn Architecture Guide: Building a High Availability and Disaster Recovery Solution by Using Failover Cluster Instances and Availability Groups
  - http://msdn.microsoft.com/en-us/library/jj215886.aspx
- AlwaysOn Solution Guide: Offloading Read-Only Workloads to Secondary Replicas
  - http://msdn.microsoft.com/en-us/library/jj542414.aspx
- Cross-cluster Migration of AlwaysOn Availability Groups for Operating System Upgrades
  - http://msdn.microsoft.com/en-us/library/jj873730.aspx
- Microsoft SQL Server AlwaysOn Solutions Guide for High Availability and Disaster Recovery
  - http://msdn.microsoft.com/en-us/library/hh781257.aspx
- Migration Guide: Migrating to AlwaysOn Availability Groups from Prior Deployments Combining Database Mirroring and Log Shipping
  - http://msdn.microsoft.com/en-us/library/jj635217.aspx
- Multisite Failover Cluster Instance
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2011/12/22/sql-server-2012-alwayson_3a00_-multisite-failover-cluster-instance.aspx
- SQL Server 2012 AlwaysOn High Availability and Disaster Recovery Design Patterns
  - http://sqlcat.com/sqlcat/b/msdnmirror/archive/2011/12/22/sql-server-2012-alwayson-high-availability-and-disaster-recovery-design-patterns.aspx

Q&A

- Domande?

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http://speakerscore.com/MK1T

THANKS!